

A Self-Consistent Spatially-Resolved Cluster Dynamics Model for Irradiated Zirconium

Nasr M. Ghoniem^{a,*}, Giacomo Po^b, Matthew Maron^b, Ruben Escobar^b, Benjamin Ramirez Flores^c,
Kristopher Baker^c, Thomas Black^d, James Hollenbeck^d

^a*Department of Mechanical and Aerospace Engineering, University of California Los Angeles, Los Angeles, CA 90095, USA*

^b*Department of Mechanical and Aerospace Engineering, University of Miami, Coral Gables, FL 33146, USA*

^c*Knolls Atomic Power Laboratory, Fluor Marine Propulsion, LLC, Niskayuna, NY 12309, USA*

^d*Bettis Atomic Power Laboratory, Fluor Marine Propulsion, LLC, West Mifflin, PA 15122, USA*

Abstract

We present a self-consistent, spatially resolved cluster-dynamics model for the irradiated microstructure of α -zirconium, together with the construction by which its continuum fields are converted into a discrete dislocation population and the two-time-scale method by which the whole is integrated. The model is self-consistent in a restrictive sense: every capture efficiency appearing in a loop growth law, and every sink strength those loops impose on the mobile species, is generated from the *same* anisotropic diffusion tensors that transport those species, through a single anisotropy parameter per species, rather than from an independent set of phenomenological bias factors. The immobile population is resolved into nine crystallographic families — the three prismatic $\langle a \rangle$ variants of both interstitial and vacancy character, and a three-state basal vacancy chain — and each carries the first three moments of its size distribution, closed by a log-normal ansatz, so that the width of the distribution is transported and not only its mean. The mobile species obey a steady reaction–diffusion problem with Dirichlet conditions at grain boundaries; the immobile families obey pointwise ordinary differential equations in which no neighbor’s state appears. A Lie–Trotter operator split exploits that structure: freezing the mobile field removes the spatial coupling by construction, so that 7.8×10^7 stiff equations are integrated as 72 928 independent 36-dimensional systems rather than one coupled system of 2.8×10^6 , a reduction of 6.5×10^9 in the cost of a single factorization. Grain boundaries are treated as sinks for both populations, by different mechanisms: a Dirichlet condition for the mobile species, and for the loops a size-gated removal of the fraction of each family’s own size distribution large enough to intersect the boundary. Applied to a 500 nm hexagonal single crystal at 573 K to 10 dpa, the model resolves mobile boundary layers of 8–46 nm that order themselves by species mobility and narrow by a factor of two as the sink density grows; closes both atom balances on production, with the boundary absorbing 41% of interstitial production as mobile species and a further 5.1% of vacancy production as loops; and predicts a loop-denuded zone ~ 5 nm wide across which the mean loop diameter recovers from 3.5 to 58 nm. Carrying the loop absorption channel removes an artifact that otherwise makes a domain average meaningless: without it, the domain and interior means of the basal loop density differ by a factor of 873. The parameters are calibrated against 78 transmission-electron-microscope measurements of loop density and diameter as a maximum *a posteriori* estimate under stated priors, and we report explicitly which of them the data determine and which rest on the bounds of the admissible range. The model produces basal loops that are too numerous by a factor of 9.5 and too small by a factor of 1.9; the two errors are linked by the number and growth balances, and we argue that they identify a missing channel that removes vacancy loop *number* without shortening the survivors’ lives.

Keywords: zirconium, cluster dynamics, spatially resolved, dislocation loops, diffusional anisotropy, grain boundary, irradiation damage

1. Introduction

Zirconium and its alloys — Zircaloy-2 and Zircaloy-4 in light-water reactor fuel cladding, Zr-2.5Nb in CANDU pressure tubes, and the low-tin variants M5 and ZIRLO in pressurized-water reactor service — operate under prolonged neutron exposure at displacement rates of 10^{-8} to 10^{-7} dpa s $^{-1}$ and accumulate total damage of 5–30 dpa over a fuel-cycle lifetime [1, 2]. The microstructural changes that result — nucleation and growth of $\langle a \rangle$ prismatic loops, the comparatively delayed appearance of $\langle c \rangle$ -component basal vacancy loops, dislocation channeling under load, anisotropic irradiation growth and creep, and stress-driven hydride reorientation — are the principal source of property degradation in Zr-based components, manifested as hardening, loss of uniform elongation, axial growth of fuel assemblies, and creep-down of pressure tubes [3, 4, 5, 6]. Predicting those changes across the joint envelope of dose, temperature, texture and applied stress is the central quantitative challenge of fuel-element and pressure-tube qualification.

Cluster dynamics (CD), in the form it took in the late 1970s and early 1980s [7, 8, 9, 10, 11], remains the workhorse mesoscopic theory for that prediction. It treats each defect species and size as a population whose concentration evolves under coupled master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission and absorption at fixed sinks [12, 13, 14]. Its advantages are direct access to time scales far beyond molecular dynamics [15, 16], explicit resolution of the cluster-size distribution and therefore of every quantity that enters a macroscopic property model [17, 18], and modest cost relative to kinetic Monte Carlo, which together make CD the meeting point between atomistic parameterization [19, 20] and engineering-scale prediction [21, 5].

1.1. The mean-field assumption, and where it fails

Classical CD is a *mean-field, spatially averaged* theory. Every concentration is a single number for the whole material, every sink is represented by a sink strength k^2 that has already been averaged over an assumed random and statistically uniform distribution of absorbers, and the transport that connects a mobile defect to a sink appears only through that average. Three distinct assumptions are folded into this, and each fails under conditions that are not exotic.

Spatial uniformity.. A sink strength is meaningful only if the diffusion field around each absorber has relaxed to a common far-field value before it meets the next absorber. Near a free surface, a grain boundary, or a denuded band, it has not: the mobile concentration varies by orders of magnitude over a distance comparable with the boundary-layer width, and a single average concentration is not merely imprecise, it does not describe any point in the material. The consequences are structural rather than quantitative. In the calculations reported below, the outermost shell of a 500 nm grain carries essentially the entire prismatic loop population, so a domain average of the loop density is a statement about the boundary layer and not about the material.

Diluteness.. The Avrami-type coalescence kernels that CD uses to represent loop–loop and loop–network interaction are derived by assuming that absorbers are placed independently and at random. Once the volume fraction swept by growing loops becomes appreciable, the placement is no longer independent — loops that have already coalesced are correlated by construction — and the mean-field estimate of the overlap probability ceases to be an estimate of anything. This is the point at which the population must be represented as discrete objects rather than as a density, and locating it is a prediction the model can make about itself.

Isotropy inherited from cubic metals.. Self-interstitial migration in α -Zr is strongly anisotropic, prism-plane mobility exceeding basal mobility by roughly an order of magnitude, and vacancy migration is anisotropic more weakly [19, 20]. Most CD treatments of zirconium nevertheless retain isotropic diffusion constants taken from the body-centered cubic literature and recover the missing directionality through independently fitted

*Corresponding author.

Email address: ghoniem@ucla.edu (Nasr M. Ghoniem)

bias factors. That conflation is not a book-keeping convenience: it allows a model to believe simultaneously that diffusion is isotropic and that its sinks are biased by the anisotropy of that same diffusion, and it severs the relation between atomic-scale parameters and predicted dimensional change.

1.2. Spatially resolved cluster dynamics

Spatially resolved cluster dynamics (SRCd) replaces the averaged concentrations by fields $c(\mathbf{x}, t)$ obeying reaction–diffusion equations on a meshed domain, so that the transport a mean-field sink strength summarizes is instead solved for [22, 23, 24]. Three things follow that a mean-field model cannot supply.

First, the boundary layer is resolved rather than parameterized. A grain boundary in mean-field CD is a number, $k_{gb}^2 \sim 6\sqrt{k^2/d}$; in SRCd it is a Dirichlet condition, and the depleted zone in front of it — whose width is set by the competition between the local diffusivity and the local sink density, and which therefore differs for each mobile species and evolves through the transient — is an output. The denuded zones seen in irradiated microstructures are a direct consequence, and they are among the few observations that speak to the transport parameters rather than to the reaction parameters.

Second, quantities that are degenerate in a mean-field model become distinguishable. Section 3.5 shows that diffusional anisotropy difference opens a window of the mobile flux ratio over which prismatic interstitial and basal vacancy loops grow together, but that the window is a condition on *one number*; a mean-field model carries that number once, while a spatially resolved model carries it as a field, and the observed coexistence of both loop characters can then be a statement about different points of the same grain rather than about a single balance holding everywhere.

Third, the fields are the natural input to a discrete model. Dislocation dynamics, micromechanical models of channeling, and fracture models all require defect positions and sizes, not densities. A spatially resolved continuum field carries the information a discrete population must reproduce — local number, local size, local orientation — and the conversion between the two becomes a well-posed problem with conservation laws attached, rather than a rescaling of a single average [25, 26, 27].

1.3. Self-consistency

The model presented here is self-consistent in a specific and restrictive sense. Every capture efficiency that appears in a loop growth law, and every sink strength that the same loops impose on the mobile species, is generated from the *same* anisotropic diffusion tensors that transport those species, through the tensorial form of the diffusional-anisotropy-difference (DAD) correction [28, 29, 30, 31]. There is no independent set of phenomenological bias factors. One consequence is immediate and is the reason the constraint is worth imposing: a change of the diffusion anisotropy now moves the growth laws and the sink strengths together, in the directions the transport dictates, and cannot be absorbed by refitting a bias.

The immobile population is correspondingly expanded, from the lumped stress-aligned pair of the earlier mean-field model to nine crystallographically resolved families: the three prismatic variants a_1, a_2, a_3 of both interstitial and vacancy character, and a three-state basal vacancy chain running from the stacking-fault pyramid through the faulted loop to the perfect loop. The expansion follows the experimental loop character [32, 33, 34, 35, 36]: prismatic $\langle a \rangle$ loops of both signs, and basal $\langle c \rangle$ -component loops that are predominantly *faulted* and that form by collapse of a pyramidal precursor rather than by accretion. Each family carries the first three moments of its size distribution, so that the width of the distribution — not only its mean — is transported, and the conversion to a discrete population can draw loop sizes from a distribution instead of assigning every loop at a point the same size.

1.4. Contributions and outline

This paper reports the formulation, its numerical solution, and its first application to a spatially resolved single crystal. Specifically:

- (i) a self-consistent SRCd formulation for α -Zr in which capture efficiencies and sink strengths are generated from the diffusion tensors (Sections 2 and 3);

- (ii) a two-time-scale operator split that makes of order 10^6 stiff ordinary differential equations per substep tractable, with the cost argument that justifies it and the splitting error that pays for it (Section 4);
- (iii) a continuum-to-discrete conversion with stated conservation guarantees, and a grain-boundary treatment that makes the boundary a sink for immobile loops as well as for mobile defects (Section 5);
- (iv) a calibration against 78 experimental measurements of loop density and diameter, presented as a maximum *a posteriori* estimate under stated priors, with an explicit account of which parameters the data determine and which the bounds determine (Section 6); and
- (v) a 500 nm hexagonal single-crystal calculation to 10 dpa reporting mobile and immobile fields, the closed atom balance including the boundary channel, the predicted denuded zone, and the discrete loop population the fields imply (Section 7).

Section 2 fixes the defect classes and the reactions among them. Section 3 gives the formulation, beginning from the general master equation and reducing it to the equations actually integrated. Section 4 is the solution method. Section 5 is the continuum-to-discrete map and the grain-boundary models. Section 6 is the calibration. Section 7 is the single-crystal study, and Section 8 draws the conclusions and states what remains open. Four appendices collect the capture-efficiency derivation, the moment closure, the peripheral-emission model that replaces fitted annealing lifetimes, and the state-vector layout.

2. Defect Classes and Reactions

2.1. The general class set

The primary irradiation defects in hcp zirconium are vacancies, self-interstitial atoms and their clusters. Because the lattice is elastically and crystallographically anisotropic, a cluster is not specified by its size alone: its Burgers-vector character and habit plane determine its contribution to irradiation growth, to hardening, and to the sink field seen by every other species. The observed loop population in irradiated Zr and its alloys is conventionally grouped into $\langle a \rangle$ loops and $\langle c \rangle$ -component loops [37, 38, 35], with

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad \mathbf{b}_c \parallel [0001], \quad (1)$$

the $\langle c \rangle$ loops carrying a Burgers-vector component along the hexagonal axis.

Four principal cluster families span the population, together with the two free monomers [39, 3, 5, 40, 41]:

$$I_n^a, \quad V_n^a, \quad S_n, \quad V_n^c, \quad (2)$$

where I_n^a is an interstitial prismatic $\langle a \rangle$ loop of n excess self-interstitials on a $\{10\bar{1}0\}$ habit; V_n^a a vacancy prismatic loop of the same habit and Burgers vector but opposite character; S_n a basal stacking-fault pyramid (SFP) of n aggregated vacancies, bounded by basal $\frac{1}{6}\langle 20\bar{2}3 \rangle$ and prismatic facets, formed directly from collapsed cascade vacancy debris; and V_n^c a basal vacancy $\langle c \rangle$ -component loop, which nucleates above a temperature- and dose-rate-dependent threshold, in part by unfaulting of supercritical SFPs, and which drives breakaway irradiation growth at high fluence [42, 43, 40]. The free monomers are the mono-interstitial C_i and the monovacancy C_v . No helium and no three-dimensional void population is carried: Zr alloys generate a negligible (n, α) helium yield, and multi-vacancy cascade debris in α -Zr collapses onto basal or prismatic habits rather than forming compact cavities [3, 5].

The classification separates *character* — interstitial or vacancy — from *crystallography* — prismatic, basal pyramidal, or basal planar — and both distinctions are load-bearing. Two families that share a habit plane and a Burgers vector but differ in character, I_n^a and V_n^a , respond to the same mobile fluxes with opposite sign; two families that share a character but differ in habit, V_n^a and V_n^c , present differently oriented capture cross-sections to an anisotropic diffusion field and therefore compete for the same vacancies on unequal terms. It is that second competition, taken up in Section 3.5, that the self-consistent formulation exists to describe without a free parameter.

2.2. The reduced set actually integrated

Equation (2) is size-resolved: each family is an infinite ladder in n . The model solved here replaces the ladder by a small number of *moments* of the size distribution per family (Section 3.7) and resolves instead the crystallographic *variants* that a size-resolved but variant-lumped model cannot distinguish. Writing the character as $s \in \{i, v\}$ and the habit as $k \in \{a_1, a_2, a_3, c_0, c_f, c_p\}$, the families carried are

$$\mathcal{F} = \underbrace{\{(i, a_1), (i, a_2), (i, a_3)\}}_{\text{prismatic interstitial}} \cup \underbrace{\{(v, a_1), (v, a_2), (v, a_3)\}}_{\text{prismatic vacancy}} \cup \underbrace{\{(v, c_0), (v, c_f), (v, c_p)\}}_{\text{basal vacancy chain}}, \quad (3)$$

nine in all, of which eight are dislocation loops and one is not: c_0 is the stacking-fault pyramid, the precursor from which the basal loops form. Write $\mathcal{F}_L = \mathcal{F} \setminus \{(v, c_0)\}$ for the eight loop families; every statement below that invokes a perimeter, a line density or a coalescence channel is a statement about $\mathcal{F}_L$.

Three features distinguish Eq. (3) from the lumped, stress-aligned pair of the earlier mean-field model, and each is a statement about crystallography rather than about the solver.

1. **The prismatic loops are resolved into their three variants a_1, a_2, a_3 , of both characters.** The variants carry their own habit normals $\hat{\mathbf{a}}_k$, so the resolved stress on each is computed rather than parameterized, and the response to an arbitrary stress state is a redistribution among the three rather than a shift of one scalar. At zero deviatoric stress they are equivalent by symmetry, receive equal shares, and summing them recovers the lumped population exactly — which is the regression test on the change.
2. **There is no basal interstitial family.** Basal interstitial $\langle c \rangle$ loops are not a reported population in neutron-irradiated Zr. A formal c_i population is retained in the growth inequalities of Section 3.5 for symmetry but is not a state variable.
3. **The basal vacancy population is a three-state chain, $c_0 \rightarrow c_f \rightarrow c_p$.** The pyramid c_0 is not planar and stores its vacancies with a nearly isotropic relaxation volume, whereas the faulted state c_f and the perfect state c_p are planar and store different vacancy content per unit area. Both transitions are thermally activated and both are one-way.

The two habits are mutually *orthogonal*: the normals $\hat{\mathbf{a}}_k$ of the three prismatic variants lie in (0001), while the basal normal is [0001] itself. That orthogonality is why no single electron-beam direction images both populations at true size — a zone axis showing one family face-on views the other edge-on — so any comparison of Eq. (3) against TEM densities must combine at least two zone axes, and a single-axis micrograph constrains at most one habit. It is also why the calibration of Section 6 treats the $\langle a \rangle$ and $\langle c \rangle$ data as separate target sets rather than as one.

2.3. Geometry: from content to radius

Each family carries a character sign η_s (+1 interstitial, −1 vacancy) and three spatially resolved fields: the *cluster density* $n_{sL}^k(\mathbf{x})$, clusters per host atom; the *defect content* $c_{sL}^k(\mathbf{x})$, point defects stored in that family per host atom; and the *second content* $q_{sL}^k(\mathbf{x})$ of Section 3.7, which carries the width of the size distribution. All three are dimensionless, so a number density per unit volume is n/Ω and a stored-defect density is c/Ω , with Ω the atomic volume.

The quantity that converts loop area into stored defects is not $|\mathbf{b}|$ but the edge component $\mathbf{b} \cdot \mathbf{n}$: a loop of radius R holds $\pi R^2(\mathbf{b} \cdot \mathbf{n})/\Omega$ defects. The mean radius, the mean size and the line density of family (s, k) therefore follow from its content and density as

$$R_{sL}^k = \lambda_k \sqrt{\frac{c_{sL}^k}{n_{sL}^k}}, \quad \lambda_k = \sqrt{\frac{\Omega}{\pi (\mathbf{b} \cdot \mathbf{n})_k}}, \quad m_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k, \quad (4)$$

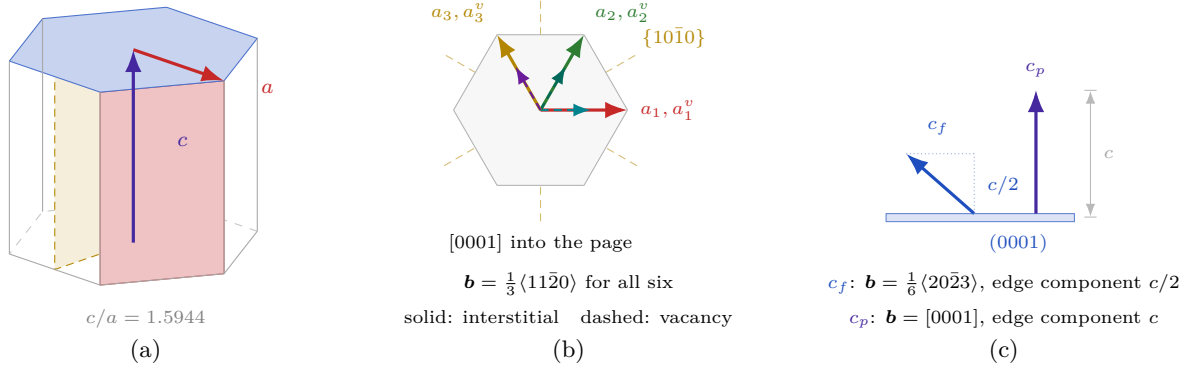


Figure 1: The hcp cell of α -Zr and every Burgers vector the model carries, drawn at the physical axial ratio $c/a = 1.5944$ rather than the ideal $\sqrt{8/3} = 1.6330$. **(a)** The cell, with the three planes that matter: the basal (0001) plane (blue), the habit of c_0 , c_f and c_p ; a first-order prismatic $\{10\bar{1}0\}$ face (red), a side face of the prism; and a second-order $\{11\bar{2}0\}$ plane (yellow, dashed), a diagonal *through* the cell rather than a face of it. The arrows are the two shortest lattice translations, $a = \frac{1}{3}\langle 11\bar{2}0 \rangle$ in the basal plane and $c = [0001]$ along the cell axis. **(b)** Plan view down $[0001]$: the three prismatic variants at 60° to one another, each drawn twice — long and solid for the interstitial family, short and dashed for its vacancy partner, which shares the habit plane and the Burgers vector and differs only in what it stores. The dashed yellow lines are the $\{10\bar{1}0\}$ traces, which bisect the Burgers directions. **(c)** The basal pair, seen edge-on. The perfect state c_p has $\mathbf{b} = [0001]$ normal to its habit, so its edge component is the full c ; the faulted state c_f has the inclined $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$, whose edge component is $c/2$ — one removed basal layer against two. That factor of two is why $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$ in Table 1, and it is why the axial ratio has to be drawn correctly. The pyramid c_0 has no Burgers vector and so appears in neither panel. The inclined pyramidal planes containing $\langle c+a \rangle = \frac{1}{3}\langle 11\bar{2}3 \rangle$ carry *no* family: that vector is the glide vector accommodating c -axis strain in deformation, and its planes are not observed habit planes for irradiation-induced loops, so the absence is an experimental statement and not a truncation.

with one geometric length per habit: λ_a shared by all six prismatic families, and λ_{c_f} , λ_{c_p} for the two basal states. Equation (4) describes a planar object and does not apply to the pyramid, whose stored vacancies fill a volume rather than bound an area; its size follows from the volumetric relation

$$R_{vL}^{c_0} = \left(\frac{m_{vL}^{c_0} \Omega}{\sqrt{8}} \right)^{1/3}, \quad (5)$$

so the pyramid carries no λ_k , no Burgers vector and no line density. What it does carry is a capture cross-section, and Section 3 shows that a cross-section is the only thing either the growth law or the sink strengths ever ask of a family. Table 1 lists the nine families with the crystallography that fixes their geometry.

Table 1: The nine families of Eq. (3), with the crystallography that fixes each one's geometry. $\lambda_k = \sqrt{\Omega/\pi(\mathbf{b} \cdot \mathbf{n})_k}$ is the length converting $\sqrt{m}$ into a radius, Eq. (4). The lattice constants are $a = 0.323$ nm, $c = 0.515$ nm, hence $c/a = 1.5944$ — the physical ratio and not the ideal $\sqrt{8/3} = 1.6330$ — and $\Omega = (\sqrt{3}/4)a^2c = 2.3266 \times 10^{-29}$ m³.

Family	Habit	$\mathbf{b}$	Character	λ_k (nm)	Comment
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	i	0.151	The dominant early population; $\mathbf{b} \cdot \mathbf{n} = a$.
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	v	0.151	Same habit and $ \mathbf{b} $ as the interstitial variants; separated from them only by capture character.
c_f	(0001)	$\frac{1}{6}\langle 20\bar{2}3 \rangle$	v	0.170	Faulted, $\mathbf{b} \cdot \mathbf{n} = c/2$, one removed basal layer. The dominant observed $\langle c \rangle$ loop.
c_p	(0001)	[0001]	v	0.120	Perfect, $\mathbf{b} \cdot \mathbf{n} = c$, two removed layers, hence $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$.
c_0	(0001) base	partials	v	—	The stacking-fault pyramid: not planar, so Eq. (4) is replaced by Eq. (5) and it carries no λ_k . Precursor to c_f .

2.4. Reaction classes

The coupled kinetics are generated from a directed multigraph — the *reaction admissibility graph* (RAG) — whose vertices are the admissible cluster states and whose edges are the elementary reactions permitted by the polarity, habit-plane and mobility constraints above. The master equations follow by a deterministic graph walk: at each vertex the time derivative is the signed sum of the rate kernels of all incoming (gain) and outgoing (loss) edges. Vertices partition by polarity,

$$\mathcal{V}^+ = \{C_i\} \cup \{C_{2i}, C_{3i}\} \cup \{(i, a_k)\}, \quad (6)$$

$$\mathcal{V}^- = \{C_v\} \cup \{(v, a_k)\} \cup \{(v, c_0), (v, c_f), (v, c_p)\}, \quad (7)$$

so the interstitial prismatic families lie on the positive branch and the six vacancy families on the negative one. Table 2 catalogs the edge classes; Fig. 2 draws the graph for the family set of Eq. (3).

Table 2: Admissible process classes of the irradiated-Zr reaction admissibility graph. Each elementary reaction maps to one abstract edge class and one rate kernel; the explicit kernels appear in Section 3 at the equation indicated. P10 has no counterpart in the mean-field treatment and is introduced in Section 5.4.

Process	Abstract edge class	Kernel	Given in
P1: vacancy–SIA recombination	cross-polarity (annihilation)	$R_{i,v}$	Eq. (11)
P2: monomer capture by loop or SFP	growth / shrinkage	ϕ_{km}	Eq. (22)
P3: cluster–cluster coalescence	same-polarity	$\varphi_{LL}, \varphi_{LN}$	Eq. (25)
P4: absorption at the network	sink edge	$R_{m,s}$	Eq. (13)
P5: thermal monomer emission	dissociation	$\mathcal{E}_{2i}, \mathcal{E}_{3i}$	Eq. (11)
P6: cascade defect production	source edge	G_m, G_k	Eq. (10)
P7: SFP $\rightarrow c_f$ collapse, $c_f \rightarrow c_p$ unfaulting	inter-population transfer	ν_{col}, ν_{uf}	Eq. (27)
P8: loop absorbed by the network	network-evolution edge	φ_{LN}	Eq. (25)
P9: absorption at a grain boundary, mobile	Dirichlet boundary	—	Eq. (15)
P10: absorption at a grain boundary, <i>loops</i>	size-gated removal	ν_{gb}	Eq. (39)

3. Cluster Dynamics Formulation

3.1. From the master equation to the reduced system

The general cluster-dynamics master equation for a population indexed by a species label α and a size n is a balance of gains and losses over the edges of the graph of Fig. 2,

$$\frac{\partial C_n^\alpha}{\partial t} = -\nabla \cdot \mathbf{J}_n^\alpha + G_n^\alpha + \sum_{\beta, n'} [\mathcal{K}_{n-n', n'}^{\alpha\beta} C_{n-n'}^\beta C_{n'}^{\beta'} - \mathcal{K}_{n, n'}^{\alpha\beta} C_n^\alpha C_{n'}^\beta] + \varepsilon_{n+1}^\alpha C_{n+1}^\alpha - \varepsilon_n^\alpha C_n^\alpha - \mathcal{D}_n^\alpha C_n^\alpha, \quad (8)$$

with $\mathbf{J}_n^\alpha = -\mathcal{D}_n^\alpha \nabla C_n^\alpha$ the diffusive flux, G_n^α the cascade source, $\mathcal{K}$ the pairwise reaction kernels, ε the thermal emission rates, and $\mathcal{D}$ the absorption rate at fixed sinks. Equation (8) is one equation per (α, n) and per mesh node, and is not solvable as written for the size ranges of interest.

Two reductions bring it to the system integrated here, and they cut in different directions.

Mobile species: truncate the ladder. Only the monomers and the smallest interstitial clusters migrate. The mobile set is closed at $\mathbf{m}_M = [-1, +1, +2, +3]^T$ — monovacancy, mono-, di- and tri-interstitial — with the signed entries m^k carrying the polarity $p^k = m^k/|m^k|$. Everything larger is immobile and is carried by the second reduction. The mobile block retains its spatial derivative, because transport is what it is for.

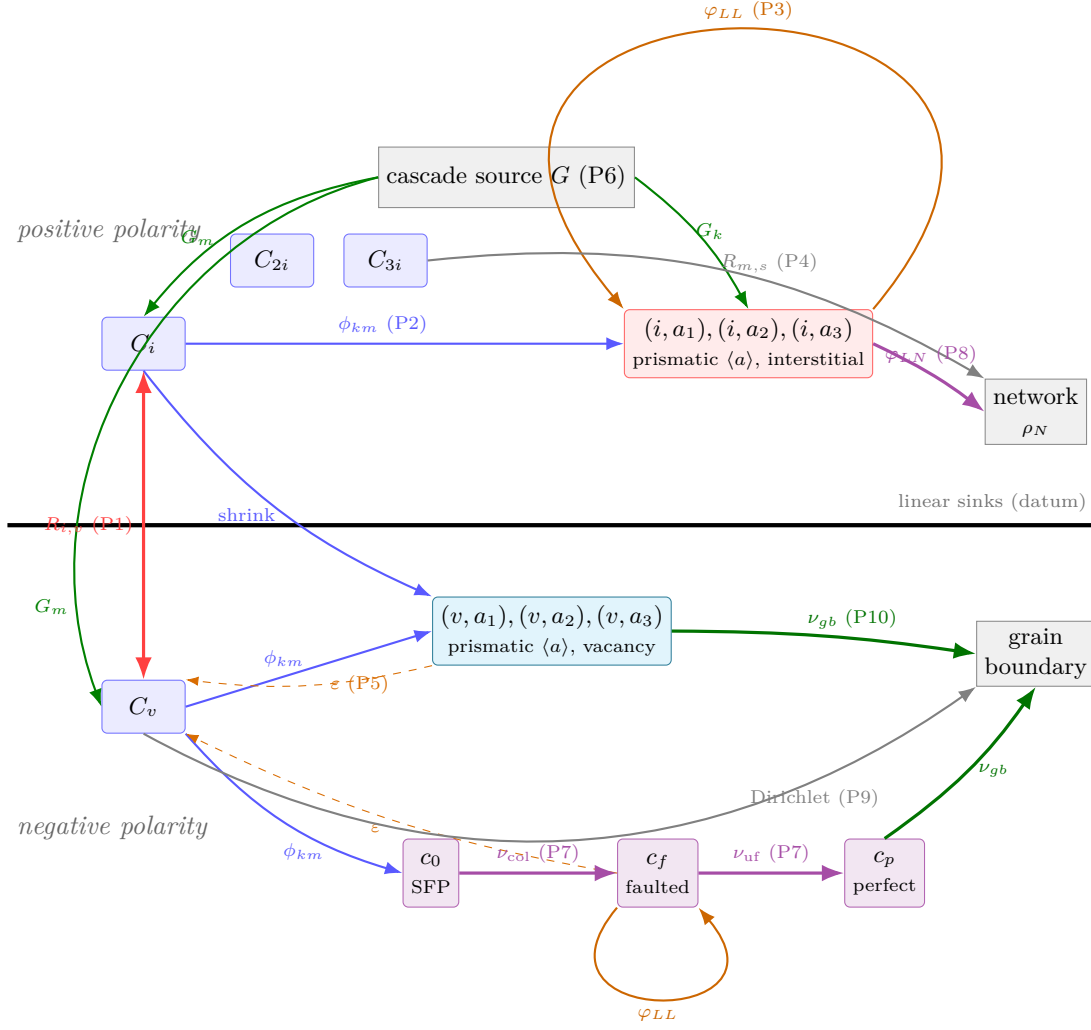


Figure 2: Reaction admissibility graph for the nine families of Eq. (3). Vertices above the datum carry positive (interstitial) polarity, those below it negative (vacancy) polarity. Blue arrows are monomer capture (P2), which acts on a vacancy loop with the opposite sign when the captured species is an interstitial; dashed orange is thermal emission (P5); the heavy red double arrow is recombination (P1); orange loops are coalescence (P3); violet is the one-way basal chain $c_0 \rightarrow c_f \rightarrow c_p$ (P7) and the loop–network transfer (P8); gray is absorption at the network (P4) and the Dirichlet condition on the mobile species at a grain boundary (P9). The green edges (P10) are the channel introduced in Section 5.4, by which a loop that has grown large enough to intersect a boundary is absorbed by it — an edge with no counterpart in a mean-field treatment, where a loop has no size relative to a boundary that is itself only a sink strength. The three prismatic variants of each character are drawn as one vertex for legibility; they are separate populations that exchange nothing with one another and differ only in habit normal.

Immobile families: truncate the moments, keep the variants.. For the immobile families the ladder in n is replaced by the first three moments of the size distribution of each family in Eq. (3),

$$n_{sL}^k = \int f_k \, dm, \quad c_{sL}^k = \int m f_k \, dm, \quad q_{sL}^k = \int m^2 f_k \, dm, \quad (9)$$

closed by the log-normal ansatz of Section 3.7. The immobile block loses its spatial derivative entirely: a loop does not diffuse, and the only thing that couples one node's loops to another's is the mobile field. That is not an approximation — it is the property that makes the numerical method of Section 4 possible.

3.2. The mobile block

Because the mobile species reach quasi-steady state on a time scale $\tau_M \sim (k^2 D)^{-1}$ of order microseconds, while the immobile families evolve on the dose scale (1 dpa = 10^7 s at $G = 10^{-7}$ dpa s $^{-1}$), the mobile block is solved as a *steady* reaction-diffusion problem for the immobile state currently held. Writing $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ in per-atom units,

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}, \quad (10)$$

where upright sans-serif bold denotes arrays in *species space* and bold italic ordinary Cartesian vectors and tensors, so that the entries $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$ of the stacked flux $\mathbf{J}$ are themselves Cartesian vectors and $\nabla \cdot$ acts on each.

The terms are, in order: the cascade production $\mathbf{P}_M = G\varepsilon \mathbf{f}$, with G the displacement rate, ε the surviving efficiency and $\mathbf{f}$ the partition of production among the mobile species; the second-order reactions $\mathbf{R}_2$, whose entries are the pairwise rate constants

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega} (\bar{D}^m + \bar{D}^n)(r^m + r^n), \quad r^m = \begin{cases} \left(\frac{3\Omega}{4\pi}\right)^{1/3}, & |m^m| = 1, \\ \left(\frac{|m^m|\Omega}{\pi b}\right)^{1/2}, & |m^m| > 1, \end{cases} \quad (11)$$

with the recombination-volume factor β_r carried on the interstitial-vacancy channel and the combinatorial factors retained on the like-species channels; the thermal emission matrix $\mathbf{E}$, whose entries are $\alpha^{m,n} = \beta^{m,n} \exp(-E_{m,n}^b/k_B T)$; the linear absorption matrix $\mathbf{D}$; and $\mathbf{S}_{vL}$, the vacancies returned to the pool by peripheral emission from vacancy loops (Section Appendix C).

The absorption matrix is where the immobile population acts back on the mobile one, and it is written so that no defect can be counted twice:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_L, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\}, \quad (12)$$

whose first term carries the *network alone*,

$$C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N, \quad (13)$$

and whose second carries the loops through the diagonal sink-strength block

$$\mathbf{K}_L = \text{diag}\{\mathbf{k}^2\}, \quad k_m^2 = \sum_{(s,k) \in \mathcal{F}_L} Z_{sk,m} \rho_{sL}^k + Z_{vc_0,m} \alpha_{\text{sfp}} \pi R_{vL}^{c_0} \frac{n_{vL}^{c_0}}{\Omega} + Z_{N,m} \rho_N. \quad (14)$$

The loops must not also appear in Eq. (13), or their absorption would be counted once as a network sink and once as loop growth. The middle term of Eq. (14) is the stacking-fault pyramid, which has no line density and enters through a compact capture cross-section instead; Eq. (24) below shows that a cross-section is the only thing either side ever asks of a family.

Boundary conditions.. On a free surface or a grain boundary the mobile species are pinned at thermal equilibrium,

$$c^v = c^{v,\text{eq}} = \exp(-E_v^f/k_B T), \quad c^i = c^{2i} = c^{3i} = 0 \quad \text{on } \partial\Omega_D, \quad (15)$$

which is process class P9 of Table 2. This replaces the mean-field grain-boundary sink strength $k_{gb}^2 \simeq 6\sqrt{k^2}/d$ by a boundary condition, and the depleted layer in front of it becomes an output rather than a parameter. Faces declared periodic instead carry periodic conditions on all four species.

3.3. Anisotropic transport and the single anisotropy parameter

Each mobile species carries a diffusion tensor $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^{xm}/k_B T)$, diagonal in the crystal frame with the basal components equal and the c -axis component distinct. The whole of that anisotropy is expressed by one number per species,

$$p^m = \left(\frac{D_{\parallel}^m}{D_{\text{basal}}^m} \right)^{1/6}, \quad (16)$$

and the migration energies are generated from it at *fixed* effective diffusivity,

$$E_{\langle 11 \rangle}^m = E_{\langle 22 \rangle}^m = E_{\text{eff}}^m + 2k_B T \ln p^m, \quad E_{\langle 33 \rangle}^m = E_{\text{eff}}^m - 4k_B T \ln p^m, \quad (17)$$

so that the orientation-averaged diffusivity $\bar{D}^m = (\det \mathbf{D}^m)^{1/3}$ is exactly the scalar D_m of the mean-field kinetics. **No mobility is redefined by the anisotropy — only its directional weighting.** This is the technical device that makes the formulation self-consistent: Eqs. (16) and (17) are a change of variable in which p^m is the only anisotropy the model has, and Section 3.4 uses that same p^m and no other parameter.

3.4. Capture efficiencies

Every capture efficiency is a product of factors, one per degeneracy to be broken:

$$Z_{sk,m}(\boldsymbol{\sigma}) = \underbrace{Z_m^0}_{\text{elastic bias}} \underbrace{A_{h(k)}(p^m)}_{\text{DAD, orientation}} \underbrace{X_{s,m}}_{\text{character}} \underbrace{\left[1 + \frac{\Delta Z_{h(k),m}^{\text{SIPA}}(\sigma_{kk})}{Z_m^0 A_{h(k)}(p^m)} \right]}_{\text{stress, SIPA}}, \quad (18)$$

with $h(k) \in \{a, c\}$ the habit of family k and the Woo diffusional-anisotropy factors [28, 29, 31]

$$A_a(p) = \frac{p + p^{-2}}{2}, \quad A_c(p) = p. \quad (19)$$

Each factor breaks exactly one degeneracy and is silent on the others. $A_{h(k)}$ depends on the habit alone — on the angle between the loop normal and the c axis — and therefore assigns one value to all six prismatic families and one to all three basal states, the pyramid included, whose base is (0001) so that $h(c_0) = c$. It separates prismatic from basal but is degenerate in character. $X_{s,m}$ is the character factor, which separates an interstitial from a vacancy loop of the same habit [44, 45]; the coexistence window it opens has log-width $\ln \chi$ with $\chi = X_{iI} X_{vV} / (X_{iV} X_{vI})$.

Equation (18) contains no free bias. $A_{h(k)}$ is built from the same p^m that Eq. (17) puts into the diffusion tensors, so a change in the transport anisotropy moves the growth laws and the sink strengths of Eq. (14) together. The two remaining constants, Z_m^0 and $X_{s,m}$, are calibrated (Section 6); the orientation dependence is not.

The forms in Eq. (19) are not monotone in the same way, and the consequence is quantitative and easily missed. $A_c(p) = p$ increases without bound, while $A_a(p) = (p + p^{-2})/2$ has its minimum at $p = 2^{1/3} = 1.2599$. A single change of p^v therefore pushes the basal and prismatic families in *opposite* directions, starving $\langle c \rangle$ growth while accelerating $\langle a \rangle$ shrinkage, or the reverse. Any statement about the effect of anisotropy that does not distinguish the two habits is incomplete.

3.5. Coexistence, and what a spatially resolved model adds

Define the mobile flux aggregates and their ratio,

$$\Phi_I = \sum_{n \in I_{\text{mob}}} n \bar{D}^{I_n} c^{I_n}, \quad \Phi_V = \bar{D}^v c^v, \quad x \equiv \frac{\Phi_V}{\Phi_I}. \quad (20)$$

The net driving force on a family of character sign η_s is $F_s = \eta_s(Z_{s,I}\Phi_I - Z_{s,V}\Phi_V)$, so that

$$\text{interstitial family grows} \iff x < \frac{Z_{s,I}}{Z_{s,V}}, \quad \text{vacancy family grows} \iff x > \frac{Z_{s,I}}{Z_{s,V}}. \quad (21)$$

Under identical local fluxes and identical capture coefficients an interstitial and a vacancy loop of the same orientation therefore *cannot* both grow. The observed coexistence of both characters is thus a statement that at least one idealization has failed.

Diffusional anisotropy resolves part of it. Because A_a and A_c differ, the prismatic and basal thresholds in Eq. (21) differ, and there is a finite window of x over which prismatic interstitial and basal vacancy loops grow together; the window has width $g(p^v)/g(p^i)$ with $g(p) = 2/(1+p^{-3})$, strictly increasing, so it is open exactly when $p^i < p^v$. But DAD is degenerate in *character*: $A_{h(k)}$ knows the habit and nothing else, so it cannot by itself support the observed coexistence of interstitial and vacancy loops on the *same* prismatic variant.

Two resolutions are carried, and the first is what a spatially resolved model supplies for free. Equation (21) is a condition on one number, and in a mean-field model that number exists once for the whole material. In SRCD, x is a *field*: it varies by orders of magnitude between the interior and the boundary layer, because c^v and c^i have different diffusivities and therefore different boundary-layer widths. Coexistence can then be a statement about different points of the same grain rather than about one balance holding everywhere. The second resolution is the character factor $X_{s,m}$ of Eq. (18), which enters as a re-parameterization of the efficiencies and not as a structural change.

3.6. The immobile block

Growth.. All nine families obey one growth law, in which the stored content changes under the net biased flux of the correct sign:

$$\dot{c}_{sL}^k \big|_{\text{growth}} = \Pi_{sL}^k \eta_s \sum_{m \in \{v,i,2i,3i\}} |m| \bar{D}^m Z_{sk,m} c^m \varsigma_s(m), \quad (22)$$

with $\varsigma_s(m) = +1$ when the polarity of species m matches the character s and -1 otherwise. The geometric prefactor is

$$\Pi_{sL}^k = \frac{\lambda_k}{\ell} \varsigma_k^{\text{sink}} \sqrt{n_{sL}^k c_{sL}^k}, \quad \ell \equiv \frac{z_c \Omega}{2\pi a^2}, \quad (23)$$

and because $\rho_{sL}^k = 2\pi\lambda_k \sqrt{n_{sL}^k c_{sL}^k}$ by Eq. (4),

$$\Pi_{sL}^k = \varsigma_k^{\text{sink}} \frac{\rho_{sL}^k}{2\pi\ell} \quad \text{identically, for every } (s,k) \in \mathcal{F}_L. \quad (24)$$

The prefactor in the growth law is the family's own sink density divided by $2\pi\ell$. That identity is what makes Eq. (22) and Eq. (14) two readings of one quantity, and it is what carries the growth law onto the pyramid unchanged: a family needs no geometry beyond the capture cross-section it presents, and the pyramid supplies that cross-section as $\Pi_{vL}^{c_0} = \varsigma_{c_0}^{\text{sink}} \alpha_{\text{sf}} R_{vL}^{c_0} n_{vL}^{c_0} / 2\ell$.

Nucleation.. Cascade production deposits directly into the immobile families at a rate $G_k = G \epsilon_k$ per family, each new cluster carrying m_k^{nuc} defects, so that $\dot{n}_{sL}^k|_{\text{nuc}} = J_{sL}^k = G_k / m_k^{\text{nuc}}$ and $\dot{c}_{sL}^k|_{\text{nuc}} = G_k$. The partition ϵ_k and the nucleation sizes are calibrated quantities.

Coalescence.. Loops are removed by overlap with other loops and with the network. With $N = n/\Omega$, $R = \lambda \sqrt{c/n}$, and the radii capped at the relevant spacing ($R_{LL} = \min(R, (\Omega/n)^{1/3})$, $R_{LN} = \min(R, \rho_N^{-1/2})$), the Avrami overlap probabilities and rate scales are

$$\begin{aligned}\varphi_{LL} &= 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi R_{LL}^3 N), & \varphi_{LN} &= 1 - \exp(-\kappa_{LN} \pi R_{LN}^2 \rho_N), \\ \nu_{LL} &= c_{LL} v_{\text{abs}} N^{1/3}, & \nu_{LN} &= c_{LN} v_{\text{abs}} \sqrt{\rho_N},\end{aligned}\tag{25}$$

giving the number and content losses

$$\Delta N_{LN} = \nu_{LN} \varphi_{LN} n, \quad \Delta N = \nu_{LL} \varphi_{LL} n + \Delta N_{LN}, \quad \Delta C = \nu_{LN} \varphi_{LN} c.\tag{26}$$

The like-loop channel merges two loops into a larger one and therefore *conserves content*, removing density only; the loop–network channel removes both, and the removed content is booked as network sink absorption so the point-defect balance stays closed. The rate scale is set by the *absorbed*-flux climb speed v_{abs} rather than the net growth velocity, so coarsening persists at saturation. The capping is what makes Eq. (25) saturate rather than diverge as the number density collapses, and $\varphi_{LL} \rightarrow 1$ is the model’s own statement that its mean-field treatment of coalescence has expired — the detector used in Section 5.1.

Emission.. Vacancy loops exchange vacancies across their own perimeter at a rate set by a size-dependent binding energy computed from the anisotropic loop self-energy (Section Appendix C), one expression serving both characters, so that the same mechanism shrinks a vacancy loop and grows an interstitial one and the classical climb threshold emerges rather than being imposed. This channel replaces the two fitted annealing lifetimes of the mean-field model. It returns vacancies to the mobile pool through $\mathbf{S}_{vL}$ in Eq. (10).

The basal chain.. The three basal states are linked by two one-way activated transfers,

$$c_0 \xrightarrow{\nu_{\text{col}}} c_f \xrightarrow{\nu_{\text{uf}}} c_p,\tag{27}$$

both size-gated, and both conserving vacancies exactly: the content leaving one state enters the next, and only the *number* changes, because the states store different vacancy content per unit area (Table 1) [46, 47, 45]. The chain is basal and has no prismatic counterpart: *ab initio*-parameterized formation energies place the prismatic vacancy $\langle a \rangle$ loop *below* both the compact cavity and the faulted basal loop, so the planar prismatic loop is already the ground-state geometry rather than something a compact cluster must convert into [47]. **None of the basal-chain rates is calibrated**; the formulation gives their Arrhenius form and no measurement in this work supplies the barriers, so the chain is switched off in the calculation of Section 7 and its consequences are discussed in Section 8.

3.7. The size distribution and its closure

Carrying only n and c assigns every loop at a point the same size, and the population that produces is monodisperse by construction. The second content q_{sL}^k removes that restriction. Its Fokker–Planck part is

$$\dot{q}_{sL}^k|_{\text{FP}} = 2\Pi_{sL}^k \left[\underbrace{\Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}}_{\text{drift}} + \underbrace{\frac{1}{2} \Sigma_k^{\text{gro}} \Delta_k^{-1/8}}_{\text{spreading}} \right],\tag{28}$$

to which nucleation, the floor current and the basal transfers are appended weighted by m^2 rather than m^1 or m^0 :

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\text{min}}^2 \Phi_{\text{min}}^k \pm \Gamma_{\theta}^q.\tag{29}$$

The spreading term does not vanish when the family stops growing: at $\Sigma_k^{\text{net}} = 0$ the drift term is zero and $\dot{q}_{sL}^k = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8} > 0$, so the distribution keeps broadening at fixed mean size and fixed number. There is no value of the two lower moments that expresses that, which is the argument for carrying the third.

Closing Eq. (29) with a log-normal $f_k(m)$ determines its two parameters from the three carried moments,

$$\bar{m}_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \Delta_k \equiv \frac{q_{sL}^k n_{sL}^k}{(c_{sL}^k)^2} = e^{s_k^2} \geq 1, \quad s_k^2 = \ln \Delta_k. \quad (30)$$

$\Delta_k \geq 1$ is Cauchy–Schwarz on a non-negative measure, so $\Delta_k < 1$ **is not a loose tolerance but a defect**: it says the three carried numbers are not the moments of any distribution. Keeping the triple realizable places a real constraint on how each channel debits q , and Section Appendix B states the three rules that follow — of which the one most easily got wrong is coalescence, because a coalescing loop does not leave the family, it *merges*, so fewer loops holding the same content are *larger* loops and coalescence *raises* q . Debiting q by the removed number times $\langle m^2 \rangle$, the natural-looking reading, inverts the sign of the largest single term in the equation.

The closure reaches the rest of the model in exactly one place, as $\Delta_k^{-1/8}$ on the sink prefactor, and it reaches the discrete conversion of Section 5 as the distribution loop sizes are drawn from.

3.8. Summary of the equations

The system integrated is two blocks with different structure, and the difference between them is the whole of the numerical method.

Mobile block — four fields, spatially coupled, no time derivative..

$$\mathbf{0} = \nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M) + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}, \quad \mathbf{c}_M|_{\partial\Omega_D} = \mathbf{c}_M^{\text{eq}}, \quad (31)$$

$4 \times N_{\text{node}}$ unknowns, nonlinear and algebraic, coupled across the mesh by the divergence term alone.

Immobile block — 27 fields, pointwise, no spatial coupling.. For each family $(s, k) \in \mathcal{F}$ at each node independently,

$$\dot{n}_{sL}^k = J_{sL}^k - \Delta N - \Phi_{\min}^k \pm \Gamma_{\theta}^n - \nu_{gb} \Theta_0^k, \quad (32)$$

$$\dot{c}_{sL}^k = \dot{c}_{sL}^k|_{\text{growth}} + G_k - \Delta C \pm \Gamma_{\theta}^c - \nu_{gb} \Theta_1^k, \quad (33)$$

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_{\theta}^q - \nu_{gb} \Theta_2^k, \quad (34)$$

with Θ_j^k the size-gated grain-boundary removal of Section 5.4 and Γ_{θ} the basal-chain transfers of Eq. (27). Together with the evolving network density ρ_N and the atom-balance accumulators of Section 7.4, this is a system of ordinary differential equations at each node — $3 \times 9 + 1$ fields plus accumulators — in which no neighbor’s state appears. **The mobile species are the only thing that couples one node to another**, and freezing them, as Section 4 does, removes the spatial coupling by construction rather than by approximation.

4. Numerical Solution by Operator Splitting

4.1. Why the coupled system cannot be integrated directly

The system of Section 3.8 on the mesh of Section 7.1 carries 2 446 659 immobile degrees of freedom and 362 468 mobile ones — 2 809 127 in all. Integrated as one coupled stiff system it would be hopeless, and it is worth being precise about why, because the numbers that follow are easy to misread as a claim that something impossible has been done.

Three properties compound. The system is *stiff*: recombination and absorption act on microseconds while the loop populations evolve over 10^7 s, so an explicit integrator is excluded and an implicit one must factorize a Jacobian. It is *large*: a dense factorization of the coupled Jacobian is $O(N^3)$ with $N = 2\,809\,127$, and sparsity softens but does not rescue that — a three-dimensional finite-element Jacobian still factorizes at $\sim O(N^2)$ with fill growing as $O(N^{4/3})$. And it is *long*: a march to 10 dpa at 10^{-7} dpa s $^{-1}$ is 10^8 s of physical time, over which an adaptive integrator takes of order 10^2 internal steps per unit of the slow dynamics.

The scale separation that makes the problem hard is also what makes it tractable. The mobile species recombine, cluster and reach the absorbing boundaries on $\tau_M \sim (k^2 D)^{-1}$, of order microseconds at 573 K and the sink densities of interest; the immobile families evolve on the dose scale. Thirteen decades separate them. On the slow manifold the mobile species are *slaved*: writing the mobile block as $\dot{\mathbf{c}}_M = \varepsilon^{-1} \mathcal{R}_M(\mathbf{c}_M; \mathbf{Q})$ and the immobile block as $\dot{\mathbf{Q}} = \mathbf{g}(\mathbf{Q}; \mathbf{c}_M)$ with $\varepsilon = \tau_M/\tau_Q \ll 1$, the singular limit $\varepsilon \rightarrow 0$ replaces the first by the algebraic constraint $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}) = \mathbf{0}$. This is the quasi-steady-state assumption, and the spatial code implements it *exactly*: its mobile solver carries no time derivative at all. Equation (31) is not a discretization of a transient problem taken to large times; it is the steady problem itself.

4.2. Structure of the split

The two blocks are advanced by a Lie–Trotter split. Over one dose interval $[\gamma_n, \gamma_{n+1}]$ with $\Delta t = (\gamma_{n+1} - \gamma_n)/G$, the scheme alternates:

- S1. Mechanics.** Solve the elastic boundary-value problem with the current inelastic deformation for $\sigma_n(\mathbf{x})$. Constant in the calculations reported here, and retained for generality.
- S2. Assemble local coefficients** from the current immobile state $\mathbf{Q}_n(\mathbf{x})$ and σ_n : the diffusion tensors, the efficiencies $Z_{sk,m}$ of Eq. (18), the sink operator of Eq. (12), the quadratic $\mathbf{R}_2$, the variant weights, and the boundary data.
- S3. Fast solve** (finite element, Newton, spatially coupled). With $\mathbf{Q}_n$ fixed, solve $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}_n) = \mathbf{0}$ for the quasi-steady mobile field.
- S4. Slow march** (pointwise, decoupled). At every node, integrate Eqs. (32) to (34) over Δt with the four mobile values frozen at $\mathbf{c}_M^*$.

The fast step must be inside the loop.. If the mobile field is instead replayed from a previously recorded calculation, the scheme is not an operator split at all: $\mathbf{c}_M$ never responds to the immobile state the march is building, so a starting state away from quasi-steady state can never relax. Any spatially uniform seed is such a state, because it carries no boundary layer while the true $\mathbf{c}_M^*(\mathbf{x})$ is pinned to thermal equilibrium on every Dirichlet face.

Solver settings must be shared across compared routes.. The spatial code’s default makes the mobile solve a *projected* Newton iteration that does not converge on this problem: it caps out at 53.6% error in c^i . The projection is disabled for every route reported here. Left at the default on one side only, a route-to-route comparison measures that solver defect rather than the physics.

4.3. The slow step, and why the mobile field must be frozen

With $\mathbf{c}_M^*(\mathbf{x})$ held fixed, the immobile equations at different nodes are *independent*. This is not an approximation introduced for speed. The only thing coupling neighboring points is diffusion of the mobile species, and freezing them removes that coupling by construction; a point’s immobile equations then depend on its own loop populations and on the four mobile values pinned into its state, and on nothing else.

The step is implemented by reusing the same stiff integrator with a flag that zeroes the four mobile derivatives *after* every reaction term has been evaluated, so the immobile equations, the conservation accumulators and ρ_N all continue to read the mobile values as frozen parameters and nothing is dropped. Each node becomes one case of an OpenMP batch integrated concurrently, with its own solver workspace, and identical states are deduplicated before dispatch — a measured factor of 1.24 at the end of the march reported here, and considerably higher early on, when much of the domain still shares one state.

The integrator is chosen by measurement.. CVODE with backward differentiation formulas and a dense linear solver, at $\text{rtol} = 10^{-6}$, $\text{atol} = 10^{-20}$. Against a tight reference, BDF with a dense or a full-band solver gives a maximum relative error of 3.0×10^{-4} ; a Krylov solver gives 1.7×10^{-2} ; and the implicit Runge–Kutta tables designed as IMEX pairs fail outright in pure implicit mode, by four to seven orders of magnitude. The frozen system is 36-dimensional, so a dense factorization is $\sim 1.6 \times 10^4$ flops and there is nothing to gain from sparsity.

4.4. Why of order 10^6 ODEs per substep is tractable

The slow step of the calculation in Section 7 integrates 3 262 212 scalar ordinary differential equations per substep and 7.8×10^7 over the march. That is *not* one system of three million equations. It is 72 928 independent systems of 36, and the distinction is worth ten orders of magnitude:

one implicit factorization	flops
split, $72\,928 \times 36^3$	3.4×10^9
coupled, $(90\,617 \times 31)^3$	2.2×10^{19}
ratio	6.5×10^9

Stiffness localizes the same way. Each point’s system genuinely is stiff — of order 100 CVODE internal steps per point per substep, which is why BDF — but stiffness cost scales with the dimension of the coupled *block*, not with how many independent blocks there are.

The coupling did not vanish; it moved to where it is affordable. The tightly coupled problem still exists: it is the fast solve, 362 468 unknowns, 4 m 12 s measured on 24 cores, against 1 m 12 s for an entire slow sweep over all 90 617 points. What the split buys is *frequency*. Because Eq. (31) has no time derivative, the coupled problem is solved eight times over the march rather than at every timestep. The counterfactual, from the same measurements at five Newton iterations per solve and hence ~ 50 s per linear solve:

$$\underbrace{100 \text{ internal steps} \times 24 \text{ substeps} \div 5 \text{ steps per refactorization}}_{\approx 480 \text{ factorizations}} \times 50 \text{ s} \approx 6.7 \text{ h},$$

for the four-species mobile block *alone*, against the 29 min the entire slow side of the march actually took.

Three things stack, then: freezing the mobile field removes the spatial coupling; the coupled part has no time derivative, so it is solved eight times rather than thousands; and batch OpenMP with deduplication spreads what is left over the available cores. The slow side is linear in node count, as this structure predicts.

4.5. Splitting error and its control

Freezing $\mathbf{c}_M^*$ over a substep ignores the drift of $\mathbf{Q}$, and hence of the sinks, within that substep. This is a first-order Lie–Trotter error, $O(\Delta\gamma)$ per step and one-sided: with strengthening sinks the loops are slightly over-grown.

The accuracy dial is the fast-solve cadence, not the dose-snapshot spacing. The march advances the immobile block in n_{sub} substeps per dose interval and re-solves the mobile field every n_{fem} substeps, so the splitting error is first order in the interval between fast solves and not in the interval between output snapshots. Two practical notes follow, both of which were established by their failure modes. The cadence counter must run over the *whole march* and not within an interval: counted within an interval, n_{fem} becomes silently inoperative whenever an interval holds fewer substeps than the cadence. And the substep count is fixed *per dose interval* rather than by a wall-clock cap on Δt : a 5 dpa interval at 10^{-7} dpa s $^{-1}$ is 5×10^7 s, which a fixed 5×10^4 s cap would have split into a thousand batch subprocesses.

Three refinements are available and are not used in the calculations reported here. A step cap accepts a step only if $\max_q \Delta \langle k^2 \rangle / \langle k^2 \rangle \leq \eta_{\text{split}}$ and halves $\Delta\gamma$ otherwise. Strang splitting — half an immobile step, a mobile re-solve, half an immobile step — gives $O(\Delta\gamma^2)$ for one extra fast solve. An in-step Picard iteration between S3 and S4 removes the splitting error entirely and is the natural fallback if a regime requires it.

4.6. Measured cost

Table 3 reports the calculation of Section 7 on 24 cores. Two readings matter. The fast and slow sides cost nearly the same, 52% against 45%, which is the balance the cadence $n_{\text{fem}} = n_{\text{sub}} = 3$ was chosen to produce: raising the cadence buys accuracy at close to linear cost in the fast side alone. And a single point-integration costs 994 μ s for 36 stiff equations over a dose interval, which is what makes a million of them per substep an hour of wall clock rather than a week.

Table 3: Measured cost of the 500 nm hexagonal single-crystal march of Section 7, on 24 cores. The state carried is 40 wide — 4 mobile, 27 immobile, 6 conservation accumulators, the network density, and the two grain-boundary accumulators of Section 5.4 — of which 36 are integrated, the four mobile values being held fixed over each substep.

Quantity	Value	
CD nodes	90 617	
mobile field, 4 species	362 468 dof	
immobile field, 9 families $\times$ 3 moments	2 446 659 dof	
elastic displacement	271 851 dof	
state carried per point	40	
integrated per point (= Newton block)	36	
substeps	24	
points integrated per substep, after deduplication	72 928 of 90 617	
deduplication factor	$\times 1.24$	
point-integrations over the march	1 750 272	
scalar ODEs integrated over the march	7.8×10^7	
fast solves (8 $\times$ finite-element Newton)	33 m 42 s	(52%)
slow substeps (24 $\times$ batch CVODE)	29 m 00 s	(45%)
checkpoint writes (32)	18.2 s	(0.5%)
march total	1 h 04 m 37 s	
one fast solve	4 m 12 s	
one slow substep	1 m 12 s	
one point-integration	994 μ s	

4.7. A point can fail, and where

The pointwise structure makes failure local, which is a diagnostic advantage worth stating. On an earlier anisotropic march on this geometry a single node failed the immobile integration at one substep, and failed *deterministically* — in the full batch, again when re-run alone, twice, across two separate invocations. It sat 51 *b* from the basal-pole face, inside the boundary layer, with entirely unremarkable concentrations: 30th to 50th percentile in every mobile species. That location is the diagnosis. Anisotropic interstitial diffusion narrows the boundary layer along the *c* axis and steepens its gradient, which puts the point beyond the integrator’s reach at $\text{rtol} = 10^{-6}$.

Retrying such a point alone distinguishes a transient from a real failure and costs a handful of cases against tens of thousands, but it does not recover a point that fails alone: that needs a solver or a model answer. The available fallback substitutes an identity step, freezing the point for that substep; it should be understood for what it is, which is declaring the march successful without that point rather than solving it. No point failed in the calculation reported here.

5. Continuum-to-Discrete Self-Consistency

5.1. Where the mean field expires, and how the model knows

Every cluster-dynamics formulation is a mean-field theory of a population of discrete objects. Its kernels are derived on the assumption that absorbers are placed independently and at random, so that the probability of overlap between a growing loop and its neighbors is a function of the volume fraction alone. That assumption is a *dilute-limit* statement, and it is standard in the rate-theory literature that it degrades as the swept fraction becomes appreciable [12, 14, 17]. Three distinct mechanisms take it apart, and they take it apart at different doses:

1. **Spatial correlation from coalescence.** Once loops have merged, their positions are no longer independent draws — a merged loop occupies the union of two former positions — so the Avrami estimate of the overlap probability stops estimating anything.

2. **Local elastic interaction.** The capture efficiencies of Eq. (18) are derived for an isolated absorber in a uniform far field. Two loops within a few radii of each other share a stress field, and the drift term that biases capture is not the sum of the two isolated ones [44, 45].
3. **Gradients.** A sink strength presumes the diffusion field has relaxed to a common far-field value between absorbers. In a boundary layer it has not, which is the failure Section 1.1 describes and which spatial resolution already repairs for the mobile species — but not for the loops, whose own kernels are still mean-field in the *local* neighborhood.

The model detects the first of these itself, and that is the useful property. The Avrami overlap fraction φ_{LL} of Eq. (25) is the model’s own estimate of the probability that a loop has met another; when it approaches unity the estimate is of order its own maximum and carries no information. The *coarsening dose*

$$d_{\text{coarsen}} = \min\{\gamma : \varphi_{LL}(\gamma) \geq \varphi^* \text{ on a fraction } \geq f^* \text{ of interior nodes}\} \quad (35)$$

is therefore not an external criterion imposed on the model but a reading of one of its own state variables, and it costs nothing to evaluate. Which family reaches it first is a property of the formulation and not of the material: in otherwise identical calculations differing only in whether the capture efficiencies are generated from the diffusion tensors or taken as fitted constants, the family that coarsens first *swaps* between the basal and the prismatic populations.

5.2. Why a discrete representation is worth having

Beyond repairing the dilute limit, the discrete representation is the interface to micromechanics. Dislocation dynamics computes the stress a population exerts and the resistance it offers to a gliding line; a channeling model needs to know which obstacles a channel has swept and which remain; a fracture model needs the local obstacle spacing. None of these takes a density as input. They take positions, radii, Burgers vectors and habit planes.

The engineering objective is therefore stronger than a one-way export. It is that the continuum and the discrete representations be *equivalent* to within stated tolerances, so that a calculation may be run in either — or in a hybrid, continuum where the population is dilute and discrete where it is not — with the choice made on grounds of dose and computational cost rather than of physics. That is what makes the conversion a problem with conservation laws attached rather than a rescaling.

5.3. The conversion and what it must preserve

The requirements cannot all be met exactly, and their ranking is a modeling decision:

1. **Stored defects, exactly.** $\sum_{\ell} m_{\ell}$ over the discrete loops of a family must equal the continuum content of the region they replace. A transfer that loses defects appears downstream as a spurious sink.
2. **Loop number, to the integer draw.** A continuum density gives a real expected count and a discrete population an integer one, so the best available is $|N_{\text{draw}} - \Lambda| \leq \frac{1}{2}$.
3. **The dispersion Δ_k , to the truncation of the size sampling.** This requirement did not exist before Section 3.7: a monodisperse family has nothing to reproduce.
4. **Sink strength, reported but not asserted on.** The line length a discrete population presents differs from ρ_{sL}^k for reasons that are physical rather than numerical — ellipticity among them — and the difference is quantified rather than forced to zero.

Only the eight loop families convert. The stacking-fault pyramid has no Burgers vector and no line, so it has no discrete representation in a dislocation code at all; it stays in the continuum across every transfer, and a conversion that folded its content into c_f would report basal vacancies as dislocation content the solver would then be free to move.

Nodal volumes.. The fields are known at nodes that are not uniformly spaced, so each node must be given the volume of material it stands for. A restricted Voronoi tessellation supplies it, restricted in the sense that sample points falling outside the crystal are rejected rather than charged to the nearest boundary node. On a hexagonal prism the bounding box is 4/3 of the crystal, so using the box would over-count the population by a third and scatter loops through six empty wedges outside the outline.

Counts.. The expected number of loops of family (s, k) at node j is

$$\lambda_{sj}^k = \frac{n_{sL}^k(\mathbf{x}_j)}{\Omega} V_j, \quad \Lambda_s^k = \sum_j \lambda_{sj}^k, \quad N_{\text{draw}} = \text{round}(\Lambda_s^k). \quad (36)$$

Both factors are needed and for different reasons: n_{sL}^k says how many loops belong in unit material, V_j how much material the node stands for. Sampling on n_{sL}^k alone would give every node equal geometric weight and over-populate a refined region in exact proportion to how well it was resolved. Rounding rather than drawing a Poisson total is deliberate: a Poisson total has $\text{Var} = \Lambda$, and a dose sweep whose loop count fluctuates by $\sqrt{\Lambda}$ between snapshots is dominated by counting noise.

Sizes.. Each loop's size is drawn from its home node's own log-normal Eq. (30), not set to the node mean. With two moments there was no distribution to draw from and every loop at a node was identical, so the exported population was monodisperse by construction — visible immediately in any size histogram. The sample is then rescaled so that $\sum_\ell m_\ell$ matches the continuum content exactly, which is requirement 1 above; the rescaling removes the rounding, and the residual is at the level of machine precision.

Geometry.. A loop of family k becomes a polygon on its habit plane, of an area matching πR^2 rather than of that circumradius — a regular n -gon inscribed in a circle of radius R has area $(n/2)R^2 \sin(2\pi/n)$, which for a hexagon is only 82.7% of the disc, and it is the stored content that must match. Its outline is elliptical with axis ratio set by the anisotropy, and its center is placed by rejection sampling inside the crystal.

What the domain refuses.. A discrete-dislocation solver silently declines a loop whose nodes leave the grain, so the conversion must predict that decision and keep the refused share in the continuum. **The share must be counted in defects, not in loops:** on one measured transfer, one refusal in six was 83% by count and 98% by stored defects. On the geometry of Section 7 the basal population becomes effectively infeasible to transfer above ~ 0.1 dpa, because a $\langle c \rangle$ loop of mean radius 25 nm centered anywhere within 25 nm of a face overhangs it; Section 7.7 reports the measured fraction.

5.4. Grain boundaries as sinks: two models, not one

A grain boundary absorbs mobile point defects and it absorbs dislocation loops, and the two require different treatments because the objects differ in kind.

Mobile species: a Dirichlet condition.. Equation (15) pins the mobile concentrations at thermal equilibrium on the boundary. Absorption is then a flux, $-\mathbf{D}^x \nabla c^x \cdot \mathbf{n}$, and the depleted layer in front of the boundary is solved for. Nothing about this is parameterized: the layer width follows from the local diffusivity and the local sink density, differs between species, and evolves through the transient.

Immobile loops: a size-gated removal.. A loop cannot be given a boundary condition, because it is not transported — it has no flux to set. What it has is a *size*, and a loop of radius $R^k = \lambda_k \sqrt{m}$ centered a distance x from the nearest face intersects that face when

$$m \geq m_{gb}(x) = \left(\frac{x}{\lambda_k} \right)^2. \quad (37)$$

The absorbed fraction of each moment is therefore the mass of the family's own size distribution above that threshold. Writing $\Phi_k^{(j)}(m_\theta)$ for the fraction of the j -th moment of Eq. (30) lying above m_θ ,

$$\Phi_k^{(j)}(m_\theta) = \frac{1}{2} \operatorname{erfc} \left(\frac{\ln m_\theta - \mu_k - j s_k^2}{s_k \sqrt{2}} \right), \quad (38)$$

the three removals of Eqs. (32) to (34) are

$$\Theta_0^k = \Phi_k^{(0)}(m_{gb}) n_{sL}^k, \quad \Theta_1^k = \Phi_k^{(1)}(m_{gb}) c_{sL}^k, \quad \Theta_2^k = \Phi_k^{(2)}(m_{gb}) q_{sL}^k, \quad (39)$$

at a rate ν_{gb} taken large, so that the touching fraction is removed essentially instantaneously and the width of the denuded zone is a *prediction* rather than a rate parameter.

Three properties of Eq. (39) are worth stating explicitly.

It is automatically realizable. What remains after removing the mass above a threshold is a genuine sub-measure of the original distribution, so $\Delta_k \geq 1$ continues to hold by construction. No channel written this way can drive the moment triple out of the region where it is the moment triple of something.

The defects do not return to the pool. This is the whole difference between Eq. (39) and the dissolution channel, which credits the freed vacancies back to $\mathbf{c}_M$. A loop swallowed by a free surface takes its defects out of the crystal, so Eq. (39) is booked to a grain-boundary ledger and to nothing else. Adding it to the dissolution term as well would return the atoms twice, once to the pool and once to the boundary — and the symptom would be a conservation residual of the wrong sign rather than an error.

The distribution must be truncated at the crystal. The log-normal has unbounded support and the crystal does not, so without a cap the far tail of Eq. (30) contains loops larger than the grain, and Eq. (37) faithfully absorbs them at every distance — reporting absorption at the center of a grain that no loop can reach. Capping the family at $m_{\max} = (R_{\max}/\lambda_k)^2$ with $R_{\max}$ the inradius replaces Eq. (38) by the mass between the threshold and the cap over the mass below the cap,

$$\tilde{\Phi}_k^{(j)}(m_\theta) = \frac{\Phi_k^{(j)}(m_\theta) - \Phi_k^{(j)}(m_{\max})}{1 - \Phi_k^{(j)}(m_{\max})}, \quad (40)$$

which reduces to $\Phi_k^{(j)}$ exactly as $m_{\max} \rightarrow \infty$. The carried moments are still inverted for (μ_k, s_k) as if untruncated, so closure and support disagree at order $\Phi_k^{(j)}(m_{\max})$; making them consistent means inverting a *truncated* log-normal from its truncated moments, which has no closed form and would put a Newton iteration inside every residual. The inconsistency is bounded by the very tail being discarded, which is the tail that has no business existing. Measured on the calculation of Section 7, the truncation leaves the near field unchanged to four digits, sets the absorption rate at $1 \mu\text{m}$ from a boundary to exactly zero where it had been finite, and raises it at 200 nm by 4% through the renormalization.

5.5. The transfer ledger

Every transfer is audited on three totals, each summed over the whole crystal regardless of which representation holds them:

$$\begin{aligned} \text{(I) loop number} \quad & N = \sum_j n_{sj}^k V_j + \#\{\text{discrete loops}\}, \\ \text{(II) stored defects} \quad & C = \sum_j \frac{c_{sj}^k}{\Omega} V_j + \sum_\ell \frac{\pi R_\ell^2 (\mathbf{b} \cdot \mathbf{n})_k}{\Omega}, \\ \text{(III) sink strength} \quad & S = \sum_j 2\pi R_{sj}^k n_{sj}^k V_j + \sum_\ell 2\pi R_\ell. \end{aligned} \quad (41)$$

(I) and (II) are conservation statements and hold to the sampling tolerance of one loop and to machine precision respectively. (III) is a *modeling* statement, and it is the one that catches mistakes: a discontinuity

in it at a transfer means the two representations disagree about what a loop of a given content is, and that is a crystallography error rather than a numerical one. Reducing the residual sink discontinuity from a factor of 4.85 to 2% was achieved by removing an inconsistency in the $\langle c \rangle$ Burgers vector between the two sides — the continuum sizing a basal loop with the full $[0001]$ where the discrete side used $\frac{1}{2}[0001]$ — and not by tuning anything.

A transfer is all-or-nothing per family. The continuum carries one mean size per family per node, so a size-selective transfer is not representable: there is no spectrum to split. More importantly, a family that is discrete *and* still present in the continuum is a sink twice over — once through its Green’s function and once through Eq. (14) — and a loop whose own sink remains in the continuum samples its own depletion field at its own line. Zeroing the transferred family removes both faults, and they are the same fault.

6. Calibration Against Experiment

6.1. Model controls and inputs

The formulation of Section 3 contains several distinct classes of input, and confusing them is the most common way to misreport a calibration. Table 4 separates them by what a change to each one *does*.

Table 4: Classes of model input, by what a change to each affects. Only the last class is calibrated in this work; the model controls select which physics is active, the crystallography is measured, and the transport is generated from a single anisotropy parameter per species.

Class	Example	What a change does
Model controls	families carried; moments carried; emission model; basal chain on/off; grain-boundary loop channel on/off	Selects which terms of Eqs. (32) to (34) are integrated at all. Changing one changes the model, not its parameters.
Crystallography	a , c , Ω , $(\mathbf{b} \cdot \mathbf{n})_k$, λ_k	Fixed by measurement (Table 1). Not free.
Transport	p^m , E_{eff}^m , ν_m , z_c	Sets both the diffusion tensors (Eq. (17)) and, through them, the orientation factors $A_{h(k)}(p^m)$ in every capture efficiency.
Elastic and character bias	Z_m^0 , $X_{s,m}$ (through χ), Z_N	The parts of Eq. (18) that the diffusion tensors do <i>not</i> generate.
Reaction and geometry constants	ε_k , m_k^{nuc} , c_{LL} , c_{LN} , κ_{LL} , κ_{LN} , $\varsigma_k^{\text{sink}}$, m_{min} , ν_{vanish} , m_{vanish}	Nucleation, coalescence, sink prefactors, and the small-size floor. This is the calibrated set.

6.2. Experimental targets

The calibration data are transmission-electron-microscope measurements of loop number density and mean diameter in neutron-irradiated Zr and its alloys, collected across twelve temperatures and four decades of dose (Table 5). They divide sharply, and the division is a constraint on what the calibration can determine.

Table 5: Experimental targets. The two families are measured under very different coverage: 71 $\langle a \rangle$ densities across twelve temperatures and four decades of dose against five $\langle c \rangle$ points at two temperatures and only above 14 dpa. **Nothing in the data constrains the basal population below 14 dpa**, while the model places resolvable $\langle c \rangle$ loops from 10^{-2} dpa; that extrapolation is unconstrained and is revisited in Section 8. The high-temperature $\langle a \rangle$ diameters ($T \geq 650$ K) are excluded because they are vacancy loops, which the reported measurements do not separate from the interstitial population.

Set	Points	T (K)	Dose (dpa)	N / d given	Sources
$\langle a \rangle$	73	473–723	0.0067–24.5	71 / 64 (44 used)	Jostsons 1977, Northwood 1974/1976/1977/1979, Gilbert 1979, Harte 2017, Kobylansky 2008
$\langle c \rangle$	5	573–583	14–35	5 / 5	Tournadre 2012, Harte 2017

6.3. The objective, and the sense in which it is a posterior mode

Let $\boldsymbol{\theta}$ be the calibrated parameters of Table 4 and $\mathbf{y}_{\text{exp}}$ the measurements of Table 5. The estimator used here minimizes

$$J(\boldsymbol{\theta}) = \sum_{f \in \{A, C\}} w_f \left[\frac{1}{|\mathcal{N}_f|} \sum_{r \in \mathcal{N}_f} \left(\log_{10} \frac{N_r(\boldsymbol{\theta})}{N_r^{\text{exp}}} \right)^2 + \frac{1}{|\mathcal{D}_f|} \sum_{r \in \mathcal{D}_f} \left(\log_{10} \frac{d_r(\boldsymbol{\theta})}{d_r^{\text{exp}}} \right)^2 \right] + J_{\text{os}}(\boldsymbol{\theta}), \quad (42)$$

with J_{os} a penalty on non-monotonic dose response. This is *not* an arbitrary least-squares choice; it is the maximum *a posteriori* estimator under a stated model of the data. Taking the measurement errors to be log-normal — which is the right family for a quantity spanning four decades and reported as an order of magnitude plus a factor — with $\log_{10} N_r^{\text{exp}} \sim \mathcal{N}(\log_{10} N_r(\boldsymbol{\theta}), \sigma_f^2)$, and taking $p(\boldsymbol{\theta})$ uniform on the bounded box of physically admissible values, Bayes’ rule gives

$$-\log p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}}) = \text{const} + \sum_{f,r} \frac{1}{2\sigma_f^2} \left(\log_{10} \frac{y_r(\boldsymbol{\theta})}{y_r^{\text{exp}}} \right)^2 - \log p(\boldsymbol{\theta}), \quad (43)$$

so that minimizing Eq. (42) over the box is exactly maximizing Eq. (43), with $w_f \propto \sigma_f^{-2}$ and $-\log p(\boldsymbol{\theta})$ flat inside the box and infinite outside. The reported parameter set is therefore a MAP point, and the weights w_f are an explicit statement about the relative credibility of the two data sets rather than a tuning knob.

The logarithmic form matters and was not the original choice. An absolute-ratio residual $(d - d^{\text{exp}})/d^{\text{exp}}$ is bounded below by -1 and unbounded above, so it penalizes a model that is a factor of two *high* four times as heavily as one that is a factor of two *low*. Against a diameter the model systematically underpredicts, that asymmetry rewards shrinking. Equation (42) is symmetric under inversion of the ratio, which is the property a ratio of two positive quantities should have.

What has not been done.. No posterior has been sampled. The full Bayesian pipeline this framing points at — a space-filling design screened by variance-based global sensitivity analysis, a Gaussian-process emulator trained on it to stand in for the solver, and Markov-chain sampling of Eq. (43) on the emulator — is tractable here, because one evaluation of Eq. (42) costs 0.2–0.5 s over the thirteen temperatures and 78 points, but it has not been run. Everything reported below is a point estimate, and no credible interval accompanies it.

6.4. The search

The parameters were fitted in stages, each stage a bounded Nelder–Mead search over a subset, with the full set polished jointly at the end: $\langle c \rangle$ nucleation and the small-size floor; the bias; $\langle c \rangle$ removal; the $\langle a \rangle$ side; then the joint polish. Staging is a practical response to a strongly anisotropic objective and not a claim about identifiability. A screening pass over the 26 candidate parameters found nine of them inert in this formulation — a change over their whole admissible range moved J by less than the convergence tolerance — and those were held fixed.

6.5. Result

Table 6 lists the calibrated set. Agreement is summarized as the mean and standard deviation of $\log_{10}(\text{model/experiment})$, where a mean of zero is unbiased and $+0.3$ is a factor of two high:

Quantity	mean	s.d.	points
$\langle a \rangle$ density	−0.208	0.574	71
$\langle a \rangle$ diameter	−0.132	0.188	44
$\langle c \rangle$ density	−0.003	0.219	5
$\langle c \rangle$ diameter	−0.001	0.096	5

Figures 3 and 4 show the fit against the measurements, and Fig. 5 the parity plot over all 78 points.

The $\langle c \rangle$ agreement is not evidence of a well-determined basal model. Five points at two temperatures, all above 14 dpa, are fitted by four parameters. The near-zero means in the last two rows say the fit reproduces those five points, which is close to a statement about the number of degrees of freedom. The $\langle a \rangle$ row is the informative one: 71 points across twelve temperatures reproduced with a scatter of a factor of $10^{0.574} = 3.7$, which is comparable to the spread among the measurements themselves.

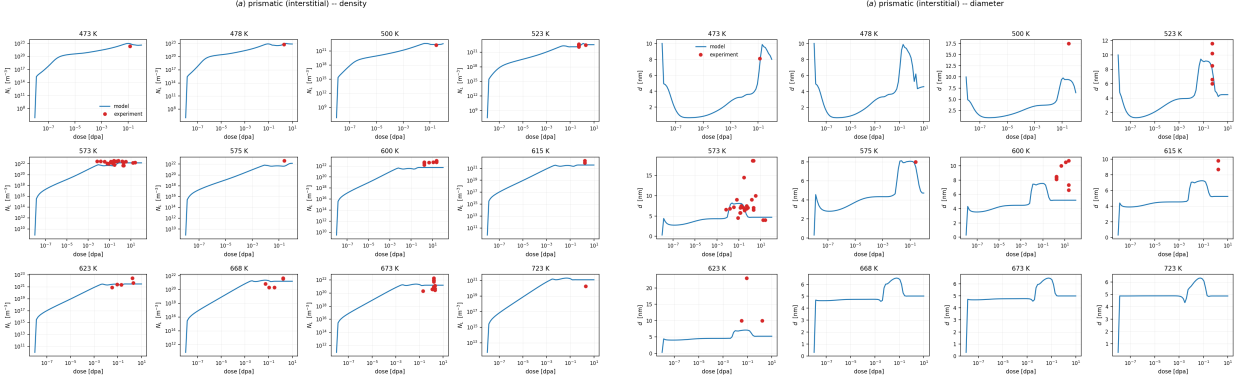


Figure 3: Prismatic $\langle a \rangle$ loops: model against experiment for number density (left) and mean diameter (right), by temperature. 71 density and 44 diameter points; the high-temperature diameters are excluded as vacancy loops.

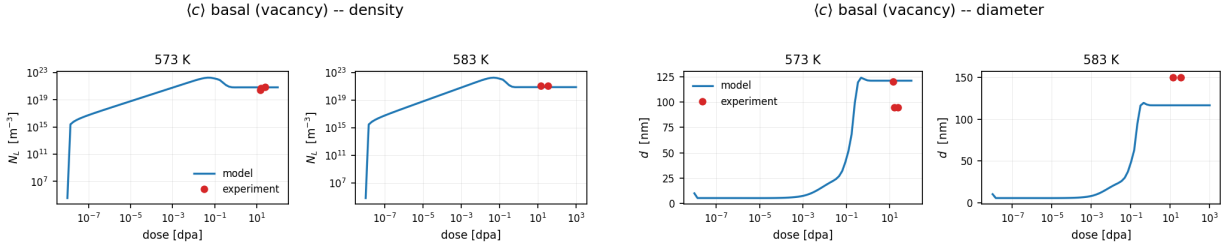


Figure 4: Basal $\langle c \rangle$ loops, against the five available measurements. Every point lies above 14 dpa; the dose axis below that carries no data, and the model's behavior there is extrapolation.

6.6. Four parameters rest on their bounds

A lever on its bound is not a fitted value. Of the calibrated set, the four bias parameters and the three small-size-floor parameters sit exactly on the edge of the admissible box, which means the data wanted to push them further and the box stopped them. The number to check is what the fitted dose anisotropy implies for the diffusion tensor. From Eq. (16),

$$\frac{D_{\parallel}^v}{D_{\text{basal}}^v} = (p^v)^6 = 16.8, \quad \frac{D_{\parallel}^i}{D_{\text{basal}}^i} = (p^i)^6 = 1.00. \quad (44)$$

That is a strong claim about vacancy transport in α -Zr, and it is not one the data in Table 5 can support. It is what the loop densities and diameters prefer, at a ceiling of $p^v = 1.6$ chosen so the implied ratio remains within an order of magnitude of what is reported atomistically; released, the search continues in the same direction. The resulting efficiencies are

$$Z_{\text{basal}}(v) = 2.080, \quad Z_{\text{basal}}(i) = 1.300, \quad Z_{\text{prism}}(i) = 1.300, \quad Z_{\text{prism}}(v) = 1.294,$$

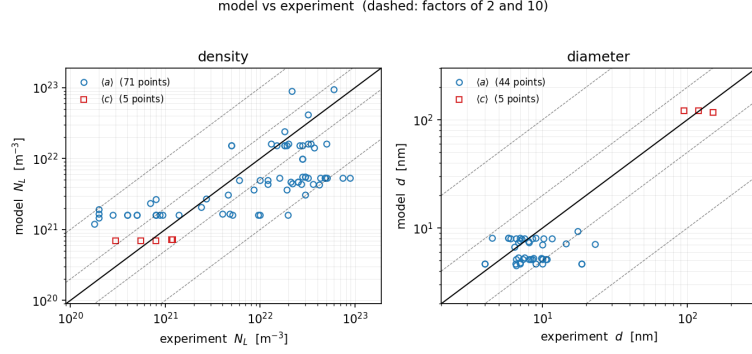


Figure 5: Parity plot over all 78 measurements. The diagonal is exact agreement and the dotted lines a factor of three.

Table 6: The calibrated parameter set. Values marked [†] rest on a bound of the admissible box and are therefore constraints rather than optima: the search wanted to go further and the box stopped it. See Section 6.6.

Parameter	Value	Parameter	Value	Parameter	Value
E_{vL}^a (eV)	0.4701	c_{LL}^a	120.58	ε_{2i}	0.0100
E_{2i}^b (eV)	0.9570	c_{LL}^c	2.0402	ε_{3i}	0.0100
E_{2i}^m (eV)	1.3629	c_{LN}^a	1130.85	ε_{iL}	0.002113
E_i^m (eV)	0.7591	c_{LN}^c	1.6720	ε_{vL}	0.006111
Q	0.2879	$c_{\rho N}$	0.0010	κ_{LL}	2.0000
Z_N	1.0100	$k_{\rho N}^{\text{rec}}$	3.5×10^{-5}	κ_{LN}	0.4000
Z_{iL}	1.3441	χ	1.0000	m_a^{nuc}	300.00
Z_{vL}	1.0200	m_{min}	1.0000 [†]	m_c^{nuc}	248.77
$Z_{v,c}$	1.1079	m_{vanish}	10.00 [†]	τ_{vL0}	1.0514
Z_v^0	1.3000 [†]	ν_{vanish}	0.0010 [†]	$\varsigma_c^{\text{sink}}$	0.7726
Z_i^0	1.3000 [†]	p^v	1.6000 [†]	p^i	1.0000 [†]

which satisfy the co-growth criterion of Section 3.5 ($p^i = 1.00 < 1.60 = p^v$) but do so at an anisotropy no independent measurement has been produced for.

Two readings follow, and they should be kept apart. As a *prediction*, the calibrated model reproduces four decades of $\langle a \rangle$ data and is used as such in Section 7. As an *inference about transport*, Eq. (44) is the strongest signal in this work that a channel is missing rather than mis-parameterized: a fit that must go to the edge of physical plausibility in one parameter to reproduce a diameter is telling us something about the model, and the diagnosis is taken up in Section 8. Independent constraints on p^v from atomistics — which Eq. (17) would accept directly, since p^m is the only anisotropy the model has — would settle it, and would turn the four [†] entries of Table 6 from constraints into fixed inputs.

Two further caveats on the transfer of this set to Section 7.. First, the fit is a 0-D fit: it is the march’s slow step term for term — one templated function, the same binary, the same switches, differing only in whether the mobile species are frozen — so the parameters transfer exactly. What does not transfer is everything the 0-D cannot see: in the 0-D the vacancy pool is free and re-equilibrates, where the march pins it to the fast solve, and the two need not move a given quantity in the same direction. Second, **the fit carries no grain-boundary loop channel**, because the 0-D has no boundary; the calculation of Section 7 does. The parameters and that calculation are therefore not strictly a matched pair, and Section 7.6 reports what the mismatch costs.

7. Results: a 500 nm Hexagonal Single Crystal to 10 dpa

Every result below comes from one calculation: the calibrated parameter set of Table 6, all nine families, three moments each, peripheral emission active, the basal chain inactive, and the grain-boundary loop channel of Section 5.4 active. Conditions are $T = 573$ K, $G = 10^{-7}$ dpa s⁻¹, zero applied stress, and eight dose intervals from 10^{-4} to 10 dpa. Where a comparison is drawn against the same case *without* the loop channel, that companion differs in nothing else.

7.1. Geometry, discretization and system size

The domain is a hexagonal prism of 500 nm across flats and 800 nm along [0001], meshed with a refined boundary layer 100 nm deep in which the target element size falls from 60 to 20 nm (Fig. 6). The refinement is not cosmetic: Section 7.2 measures interstitial boundary layers of 16–28 nm, so an unrefined mesh at 60 nm would resolve the steepest gradient in the problem with a single element.

The cluster-dynamics trial functions live on second-order elements, so the field nodes are more numerous than the mesh vertices: 90 617 of them, filling a crystal of 1.295×10^8 nm³, or 5.568×10^9 atoms. The resulting system sizes are those of Table 3: 362 468 mobile degrees of freedom solved as one coupled algebraic system eight times, and 2 446 659 immobile ones marched as 72 928 independent 36-dimensional stiff systems at each of 24 substeps — 7.8×10^7 scalar ordinary differential equations over the march, in 1 h 04 m on 24 cores.

The whole surface is a boundary. No face is periodic, so every face carries Eq. (15), and the maximum distance from any node to the nearest face is 215 nm — the inradius, and the cap $R_{\max}$ used in Eq. (40).

7.2. Mobile defect fields

Figure 7 shows the four mobile species early in the transient (10^{-4} dpa) and at the end of the march (10 dpa), on two orthogonal mid-cuts. Every panel has the same structure — an interior plateau and a depleted shell — and the differences between them are entirely in the width of that shell and in the level of the plateau.

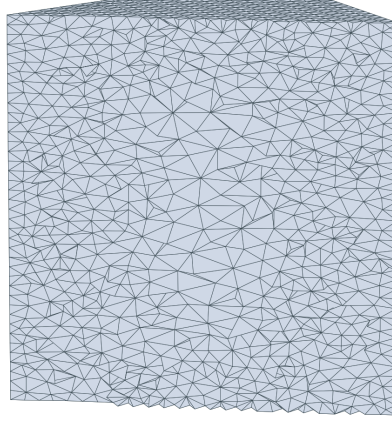


Figure 6: The finite-element mesh, cut in half so that the interior is visible: 60 922 tetrahedra on 12 581 mesh vertices. The refinement toward the six prismatic faces and the two basal faces is the boundary layer, 100 nm deep, within which the element size falls from 60 to 20 nm. Crystal axes as in Fig. 1: $[10\bar{1}0]$, $[\bar{2}\bar{1}\bar{1}0]$ and $[0001]$.

The boundary layer is a diffusion length and it orders the species.. Measuring the layer as the distance at which the binned median recovers $1 - e^{-1}$ of the interior plateau:

Dose (dpa)	C_v	C_i	C_{2i}	C_{3i}
10^{-4}	16.5 nm	27.5 nm	46.0 nm	46.0 nm
10^{-2}	14.9 nm	27.5 nm	46.0 nm	46.0 nm
10	8.0 nm	16.5 nm	18.2 nm	18.2 nm

Two readings, and both are consequences of $L \sim \sqrt{D/k^2}$ rather than of anything fitted. *Across species at fixed dose*, the layer widens with mobility: the vacancy, least mobile, is depleted over the shortest distance, while the interstitial clusters — which reach a sink from further away — carry the widest layer. *Across dose at fixed species*, every layer narrows, by a factor of two between 10^{-4} and 10 dpa, because k^2 has grown with the loop population that the same calculation is building. The boundary layer is therefore not a fixed geometric feature of the domain; it is a running measure of the sink density, and it tightens as the microstructure develops.

Anisotropy acts on the shape of the layer, not only its width.. Because the domain has two basal faces and six prismatic ones, the layer can be measured separately toward each set, using only the nodes for which that set is the nearer boundary:

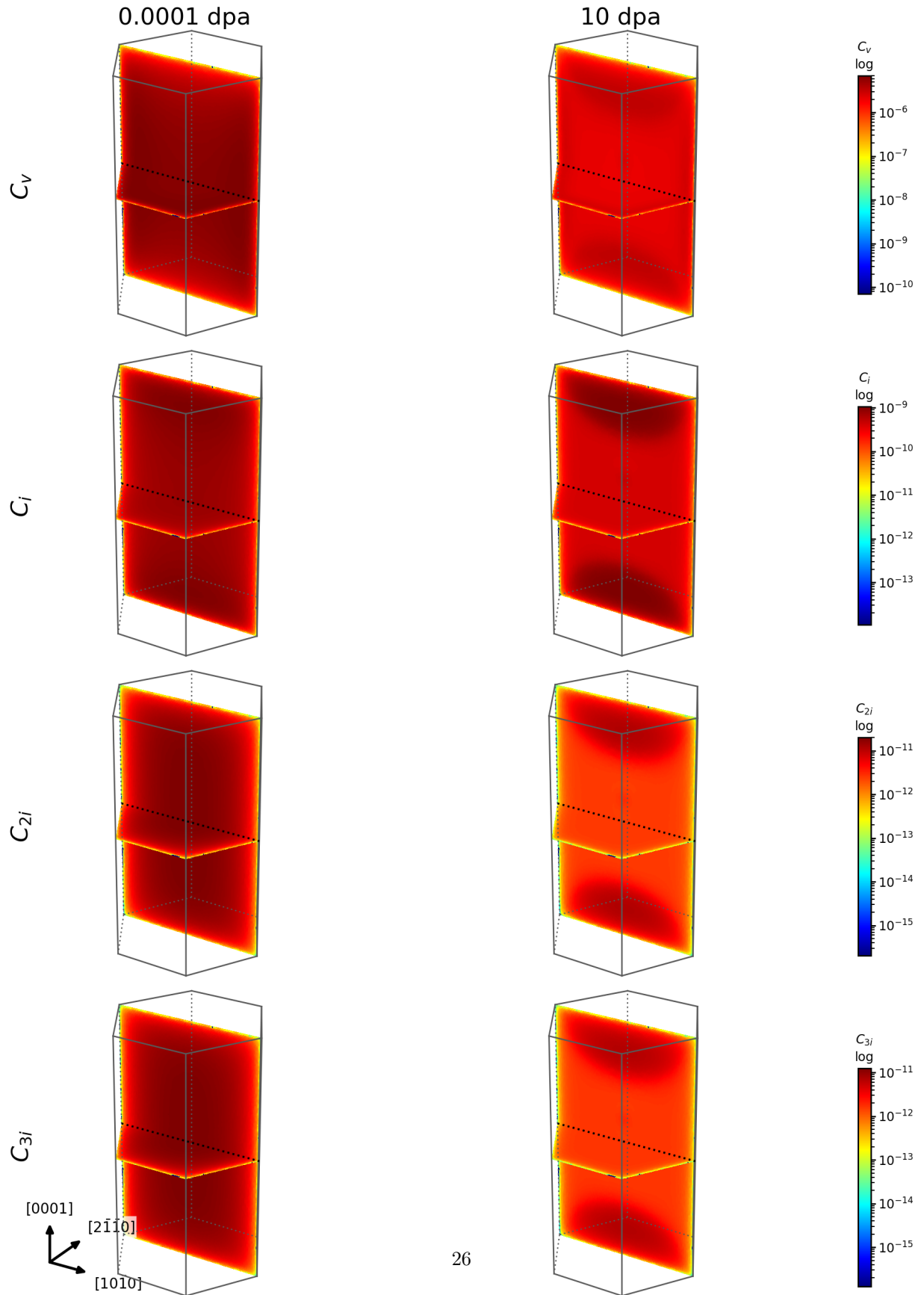


Figure 7: Mobile species at 10^{-4} dpa (left) and 10 dpa (right), on one vertical and one horizontal mid-cut. Rows, top to bottom: C_v , C_i , C_{2i} , C_{3i} ; each row has its own logarithmic color scale, so levels are comparable across dose within a row and not between rows. The interior plateau falls and the depleted shell narrows between the two doses, both because the sink density has risen.

Species	10 ⁻⁴ dpa		10 dpa	
	toward basal	toward prismatic	toward basal	toward prismatic
C_v	84.8 nm	11.7 nm	24.0 nm	6.8 nm
C_i	12.8 nm	28.8 nm	7.5 nm	18.3 nm
C_{2i}	22.0 nm	45.2 nm	6.8 nm	24.0 nm
C_{3i}	24.0 nm	45.2 nm	6.8 nm	24.0 nm

The vacancy layer is 7.2 times *wider* toward the basal faces than toward the prismatic ones, which is the sign the fitted $p^v = 1.6$ requires: vacancies are 16.8 times more mobile along [0001] (Eq. (44)), and a species that reaches a sink from further away is depleted over a longer distance. The scale expected from $L \sim \sqrt{D/k^2}$ is $\sqrt{16.8} = 4.1$, so the measured 7.2 is not a clean reading of the diffusivity ratio, and the interstitial rows show why. At $p^i = 1.0$ the interstitials diffuse isotropically, yet their layer is *narrower* toward the basal faces by a factor of 0.44 — an effect of domain shape rather than of transport, since a node near a prismatic face is close to several faces at once and is depleted more than its distance to the nearest one would suggest. Anisotropy and geometry therefore act in opposite directions here, and only their combination is directly observable.

Two consequences. First, an SRCD calculation on a shaped grain measures the convolution of the diffusion tensor with the grain geometry, so extracting one requires the other to be known — which is an argument for calibrating p^m atomistically rather than from a boundary-layer profile (Section 8). Second, the narrowest layer in the problem is the interstitial one toward a basal face, 7.5 nm at 10 dpa, against a 20 nm boundary element: the steepest gradient the calculation contains is resolved by well under one element, and it is at exactly such a location that an earlier calculation on this geometry drove a single node beyond the integrator’s reach (Section 4.7).

Interior median concentrations fall over the march — C_v from 6.02×10^{-6} to 2.29×10^{-6} , C_i from 7.92×10^{-10} to 4.16×10^{-10} , and the clusters by nearly an order of magnitude — which is the signature of a rising sink density at constant production.

7.3. Immobile fields and their moments

Figures 8 and 9 show the three carried moments of the faulted basal family and of one prismatic interstitial variant, at the same two doses. The three rows are not three views of one thing; they answer different questions, and their spatial structures differ.

Number is boundary-peaked, content is interior-peaked, and that is the physics.. Cascade nucleation is spatially uniform, but the only channel that removes a loop in the interior is coalescence, whose rate is driven by the absorbed mobile flux — and that flux vanishes where the Dirichlet condition pins the mobile species. Near a boundary, loops are therefore nucleated and not removed, so n rises steeply toward the surface. They are also not *fed*, because the same vanishing flux is what would grow them, so c does not rise with n and the mean size c/n collapses. Without the channel of Section 5.4 that shell dominates every domain average, which is the artifact Section 7.6 quantifies.

The second moment does work, and it is realizable everywhere.. The dispersion $\Delta_k = qn/c^2$ of Eq. (30) must be ≥ 1 for the three carried numbers to be the moments of any distribution at all. Measured over the interior:

Dose (dpa)	c_f		a_1		a_1^v	
	median	min	median	min	median	min
10 ⁻⁴	1.003	1.001	1.002	1.001	1.001	1.001
10 ⁻²	1.473	1.098	1.883	1.090	1.509	1.076
1	1.559	1.250	2.724	2.442	2.215	2.112
10	1.585	1.266	2.705	2.439	2.175	2.104

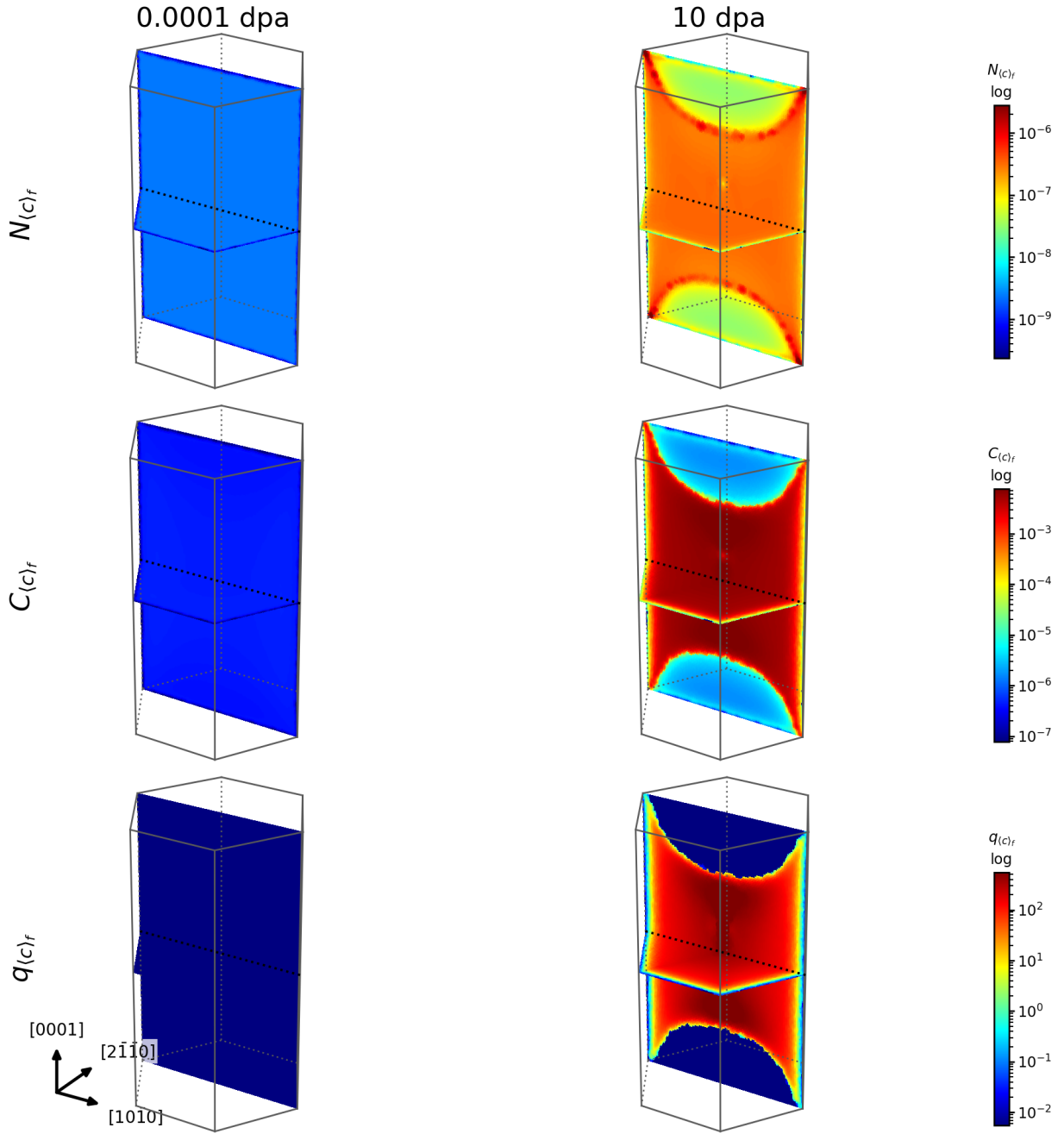


Figure 8: The basal faulted family c_f at 10^{-4} and 10 dpa : number density n (top), stored content c (middle), second content q (bottom).

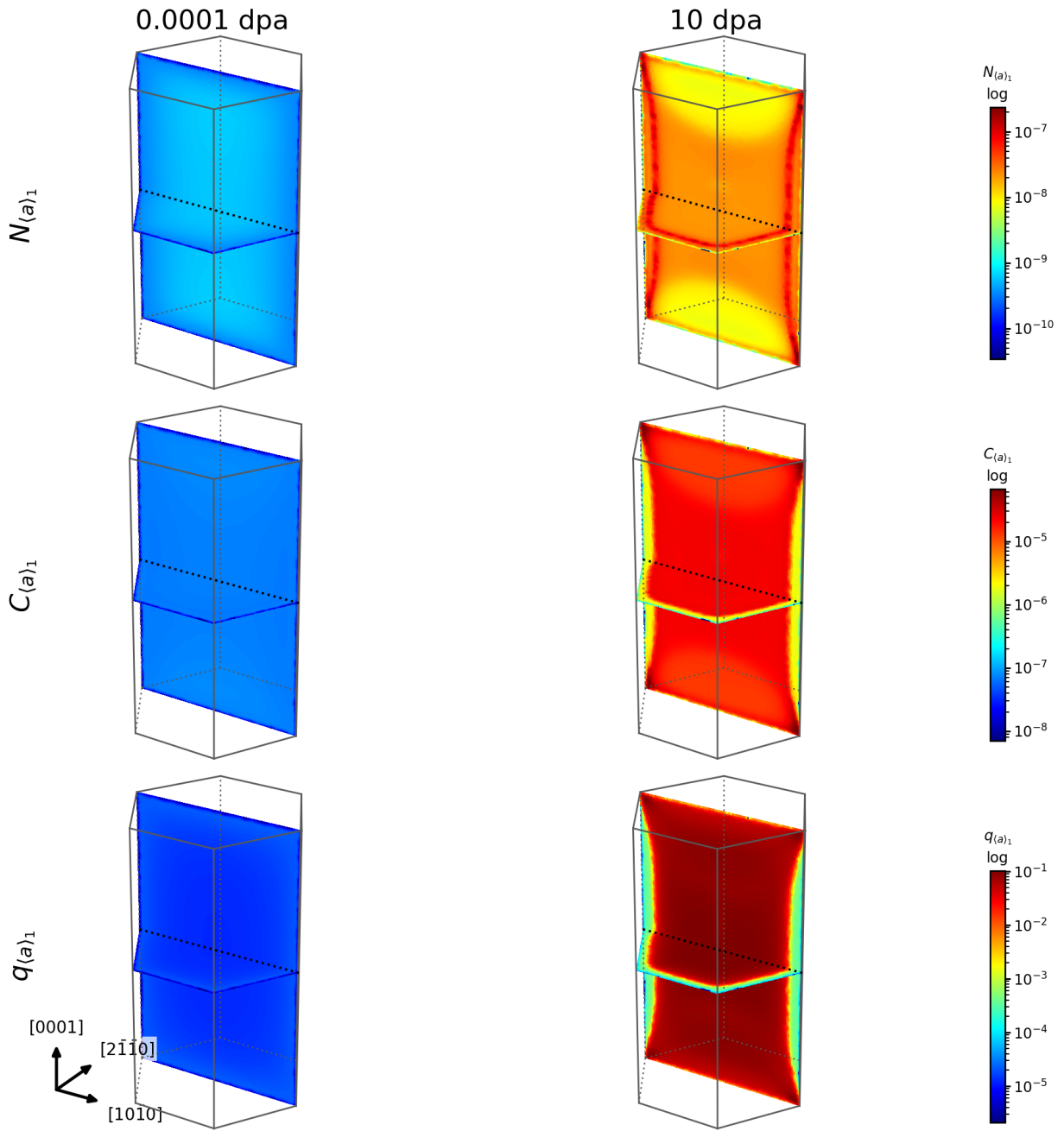


Figure 9: The prismatic interstitial variant a_1 at the same two doses, same three moments. The variants a_2 and a_3 are identical at zero applied stress and are not shown.

Three things follow. At 10^{-4} dpa every family is monodisperse to three decimal places, as it must be — the loops present are all fresh nuclei deposited at m_k^{nuc} . By 1 dpa the prismatic distribution has broadened to $\Delta = 2.72$, i.e. $s = \sqrt{\ln \Delta} = 1.00$, a log-normal a full decade wide at $\pm 1\sigma$: a model carrying only n and c would represent that population by a single size. And $\Delta \geq 1$ holds at *every* interior node at every dose, minimum 1.001, which is the check that the channels debiting q are consistent with those debiting n and c — a check that no tolerance can substitute for, since $\Delta < 1$ is not an inaccuracy but a statement that the state is not a moment triple.

The basal family broadens less than the prismatic one (1.59 against 2.71) because its number loss is dominated by the boundary channel, which removes the large end and therefore *narrows* what remains, while the prismatic family broadens under growth with no comparable truncation.

7.4. Global conservation

The six atom-balance accumulators integrate the production, recombination and sink absorption of each character, and the two accumulators of Section 5.4 integrate the loop content taken by the boundary. In a spatially resolved calculation on a Dirichlet domain the balance cannot close on those channels alone, and the residual is not solver error: it is the atoms that left through the surface as *mobile* species. The magnitude settles the matter — nothing in an integration at $\text{rtol} = 10^{-6}$ is tens of percent wrong.

Table 7 is the balance over the whole march, per host atom.

Table 7: The atom balance over $0 \rightarrow 10$ dpa, volume-averaged, per host atom. Production and recombination are exactly equal between the characters, as Frenkel-pair production and recombination must be. The two channels that differ are where the physics is: the network absorbs more interstitials than vacancies (the bias), and the boundary absorbs vacancy *loops* 60 times more than interstitial ones.

Channel	interstitial	vacancy	% prod (i)	% prod (v)
produced	10.0000	10.0000	100.00	100.00
recombined	5.0832	5.0832	50.83	50.83
absorbed at the network	0.7885	0.5908	7.88	5.91
loops at the boundary	0.0085	0.5123	0.08	5.12
stored in the microstructure	0.0000	0.0020	0.00	0.02
mobile flux to the boundary (by difference)	4.1198	3.8117	41.20	38.12

Three readings.

Both columns close on production. $10.0000 = 5.0832 + 0.7885 + 0.0085 + 0.0000 + 4.1198$ and likewise for the vacancies, so once the loop channel is carried explicitly, what remains in the residual is the mobile flux alone and the two do not contaminate each other. Before that channel was booked separately, its 5.1% of vacancy production was silently absorbed into the residual, and because the residual is closure by difference, nothing looked wrong.

The boundary is the dominant sink, by a wide margin. It takes 41% and 38% of production as mobile species, against 7.9% and 5.9% at the network. That is a property of the geometry — a 500 nm grain with every face absorbing — and it is precisely what a mean-field model represents by a single k_{gb}^2 and cannot resolve.

The loop channel is strongly asymmetric. It removes 0.512 vacancies per atom against 0.0085 interstitials, a factor of 60. The asymmetry is geometric, not biased: $\langle c \rangle$ loops reach a mean radius of 25 nm where $\langle a \rangle$ loops reach 3 nm, so a basal loop intersects a face from far deeper inside the crystal. Note also that the boundary removes *250 times more* vacancy loop content than the microstructure retains (0.512 against 0.0020), which is the quantitative form of the statement that on this grain size the loop population is boundary-controlled.

Figure 10 shows the channels against dose and the fractional decomposition. The fractions sum to unity once the boundary channels are included; without them the sum reached only ~ 0.3 .

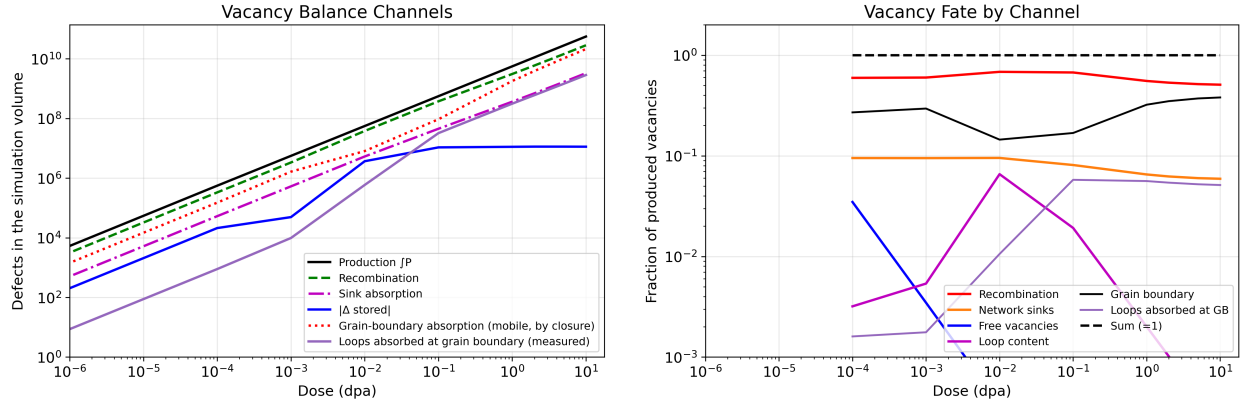


Figure 10: Vacancy balance against dose: cumulative channels (left) and their fractions of production (right). The interstitial counterparts are similar except that the loop-at-boundary channel is 60 times smaller.

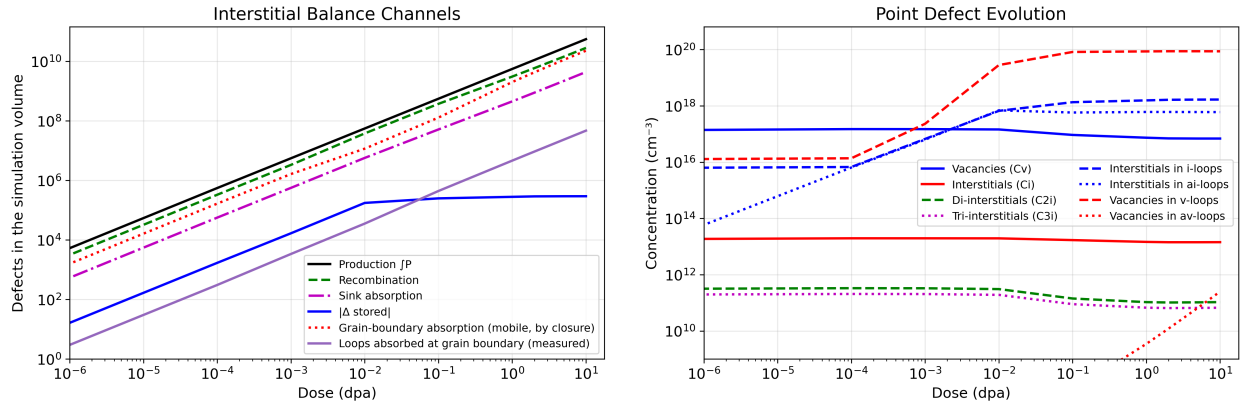


Figure 11: Left: the interstitial balance against dose, the counterpart of Fig. 10. Right: the volume-averaged mobile concentrations, whose fall through the march is the rising sink density of Section 7.2 seen as an average rather than as a field.

7.5. Grain-boundary effects and the denuded zone

Figure 12 is the central result of the loop channel: loop density and mean diameter against distance to the nearest face, with and without it.

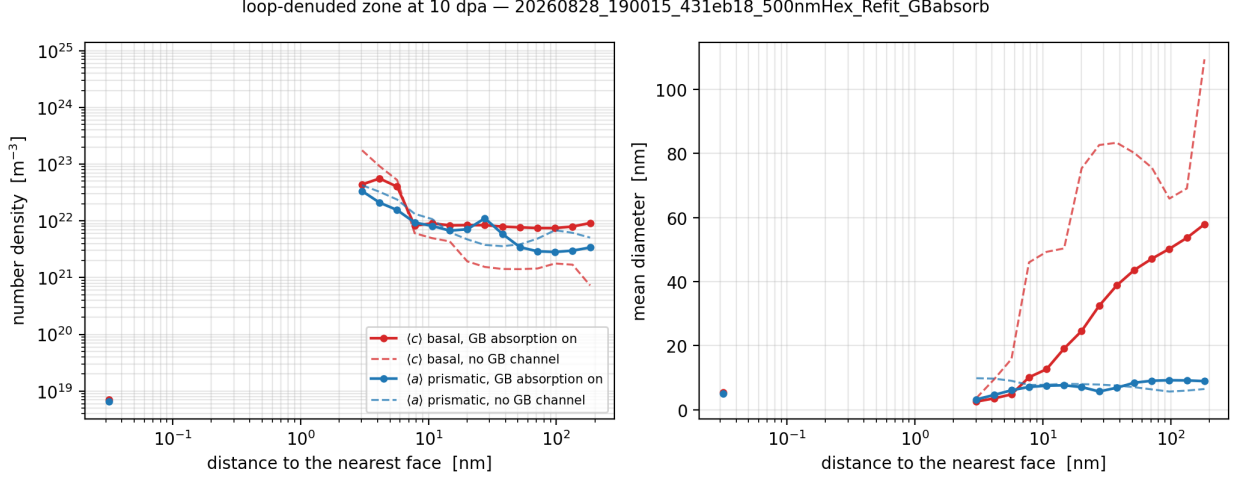


Figure 12: The loop-denuded zone at 10 dpa. Solid: grain-boundary loop absorption active. Dashed: the identical calculation without it. Left, number density; right, content-weighted mean diameter. The abscissa is the minimum distance to any face, and the ordinate on the left is logarithmic over six decades.

x (nm)	N_a/twin	N_c/twin	d_a (nm)	d_c (nm)
0–1	10^{-6}	10^{-6}	5.0	5.4
3.5–4.9	0.646	0.608	4.6	3.5
6.7–9.1	0.703	1.36	7.1	10.1
23.6–32.3	2.95	5.45	5.7	32.5
83.3–114.3	0.415	4.21	9.2	50.2
156.8–215	0.670	12.6	8.9	57.9

The zone is real, sharp, and about 5 nm wide. The face layer is swept to 10^{-6} of the companion calculation — which is the concentration floor, so as complete a removal as this model can express — and the depletion is still measurable at 5–9 nm. Beyond that the profile is no longer denuded. The mean diameter recovers monotonically from 3.5 nm at the boundary to 58 nm in the interior, which is the size gradient a denuded zone is identified by in a micrograph: not merely fewer loops near the boundary, but *smaller* ones, because the large end of the distribution is what Eq. (39) removes first.

The ledger and the density profile measure different things, and both are correct. Of the total absorption, only 1.7% occurs within 2 nm of a face and 93% between 5 and 100 nm. That is not a contradiction of the profile above: the face layer is swept within the first dose interval, so for 99% of the march there is nothing left there to absorb, while the region further in keeps *supplying* loops that grow into contact. The density says where loops are absent; the ledger says where the flux into the boundary came from over the whole history.

A local channel with non-local consequences. The interior $\langle c \rangle$ density is 4–5 times the companion calculation’s at *every* distance, including 200 nm from any face where Eq. (37) removes nothing. The cause is the operator split working as intended: removing the shell’s loops removes their sink strength from Eq. (14), the fast solve returns a different mobile field everywhere, and the interior responds to it. A model in which the boundary were only a mean-field sink strength could not produce this, because the boundary’s effect on the interior would be a single number rather than a changed field.

Comparison with experiment.. Loop-denuded and void-denuded zones at grain boundaries are a standard observation in irradiated metals [48, 49, 50, 51], and the mechanism assumed here — the boundary as a strong sink whose influence extends a diffusion length into the grain — is the standard interpretation. **We do not compare quantitatively against a measured zone width in zirconium:** no such measurement is in the data set of Table 5, and the width predicted here (a few nm for $\langle a \rangle$, growing to tens of nm in the size profile) is offered as a prediction rather than as a validated result. It is also, at 2–5 nm for the prismatic family, comparable with the 20 nm element size of the boundary layer, so the $\langle a \rangle$ zone in particular is under-resolved and its width should be treated as an upper bound set by the mesh.

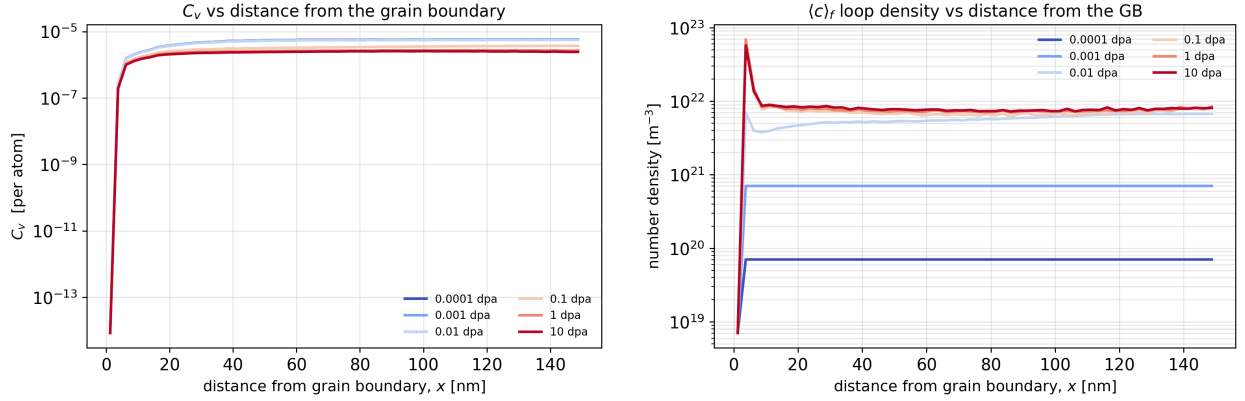


Figure 13: Profiles against distance from the boundary over the first 150 nm, with every dose overlaid. Left: the mobile vacancy concentration, whose recovery length is the boundary layer tabulated above and which narrows as the dose accumulates. Right: the basal $\langle c \rangle$ loop number density, whose profile is the denuded zone of Fig. 12 seen at every dose rather than only at the last.

7.6. Volume averages and comparison with experiment

Figure 14 shows the volume-averaged trajectories.

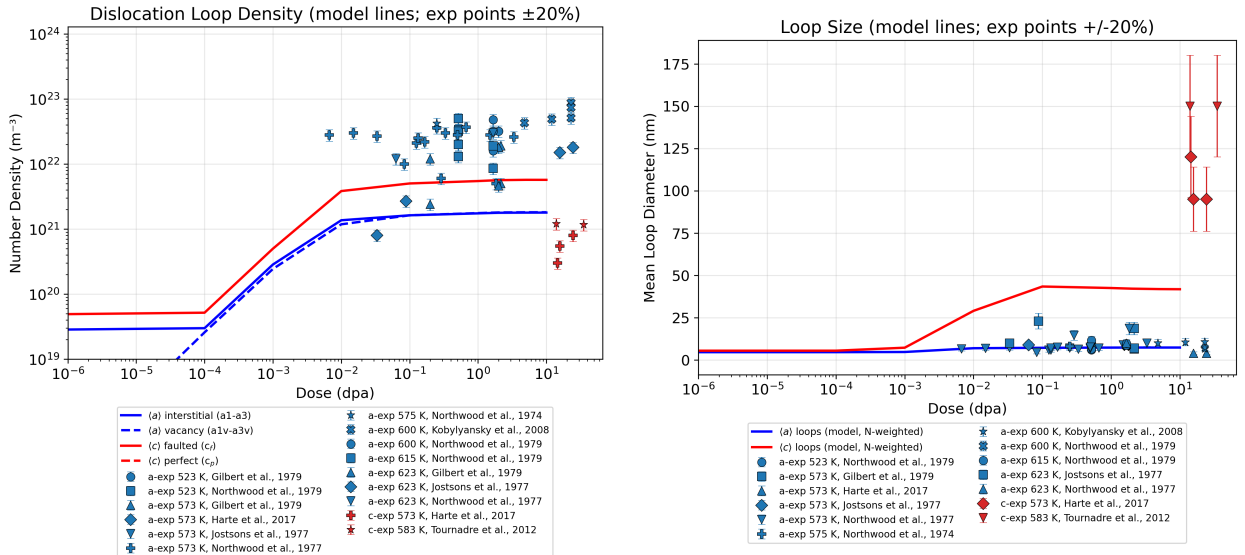


Figure 14: Volume-averaged loop density against dose (left) and mean size against the experimental measurements (right).

Which average is quoted is not a presentational choice. On a Dirichlet domain the boundary shell is a sink for mobile defects and — without the channel of Section 5.4 — not a sink for loops, so

it accumulates a loop population that is numerous, tiny, and entirely unphysical as a description of the material. At 10 dpa:

	Region	N_a (m ⁻³)	d_a (nm)	N_c (m ⁻³)	d_c (nm)
loop channel off	domain	1.32×10^{24}	5.01	1.41×10^{24}	5.74
	interior	6.06×10^{21}	5.93	1.62×10^{21}	70.16
loop channel on	domain	4.61×10^{21}	7.41	6.74×10^{21}	37.07
	interior	2.90×10^{21}	9.16	7.64×10^{21}	50.62

Without the channel the domain and interior means differ by a factor of 217 in N_a and 873 in N_c , and the domain-mean diameter *falls* to 5.7 nm while the interior mean grows to 70 nm — the domain average is measuring the boundary shell and nothing else. **With the channel the two agree to within a factor of 1.6**, and a domain average becomes a statement about the material. That is the strongest single argument for carrying the channel: it removes an artifact that otherwise makes the most natural summary statistic meaningless.

Against experiment, and the discrepancy.. At 573 K and 24.5 dpa the measurements give $N_c = 8.0 \times 10^{20}$ m⁻³ and $d_c = 95$ nm. The interior of this calculation at 10 dpa gives $N_c/N^{\text{exp}} = 9.5$ and $d_c/d^{\text{exp}} = 0.53$: **too many loops, and too small**. The companion without the loop channel gives 2.0 and 0.74 — closer on both, which is expected and is not an argument against the channel: the parameters of Table 6 were fitted in a 0-D that has no boundary at all (Section 6.6), so the channel is an unmodeled loss acting on a population calibrated without it. A refit with the channel active is required before either number is quoted as a prediction.

The direction of the discrepancy is the informative part, and it is the same one the calibration reported. The model produces *more, smaller* $\langle c \rangle$ loops than are observed, and the two errors are linked: the number balance pins the total content while the growth balance pins the size, so density is their quotient and a model that removes loops faster than it grows them necessarily has survivors that are too young. What the data want is *fewer and larger*, and no channel exercised here achieves both. Section 8 states what this implies.

7.7. Continuum-to-discrete conversion

The conversion of Section 5.3 was applied to the fields at four doses. The number of loops is not a parameter of the conversion: it is $\sum_j n_j V_j$, the number the field says is present.

Family	10 ⁻⁴ dpa	10 ⁻² dpa	1 dpa	10 dpa
c_f (basal, vacancy)	9	858	1031	1073
a_1 (prismatic, interstitial)	2	132	67	73
a_1^v (prismatic, vacancy)	2	87	56	61

Figure 15 shows the basal population at those four doses, drawn at true radius and true position, over the continuum content field it was drawn from. The two representations agree loop for loop: the figure is built from the same draw the conversion exports, so a reader comparing this figure with the exported loop table is comparing one object with itself.

Three observations.

The count is not monotone, and it should not be. Between 10⁻² and 1 dpa the prismatic count *halves*, from 132 to 67, while the stored content rises. That is coalescence: loops merge, so fewer, larger loops hold more content. A conversion that preserved number rather than content would have hidden this.

A quarter of the basal loops do not fit. At 10 dpa, 287 of 1073 $\langle c \rangle$ loops (26.7%) are centered within their own radius of a face and overhang it, by up to 40 nm; the corresponding prismatic figure is 8.2%. The drawn population is clipped at the surface, but the clip is a drawing convention and the exported loops are

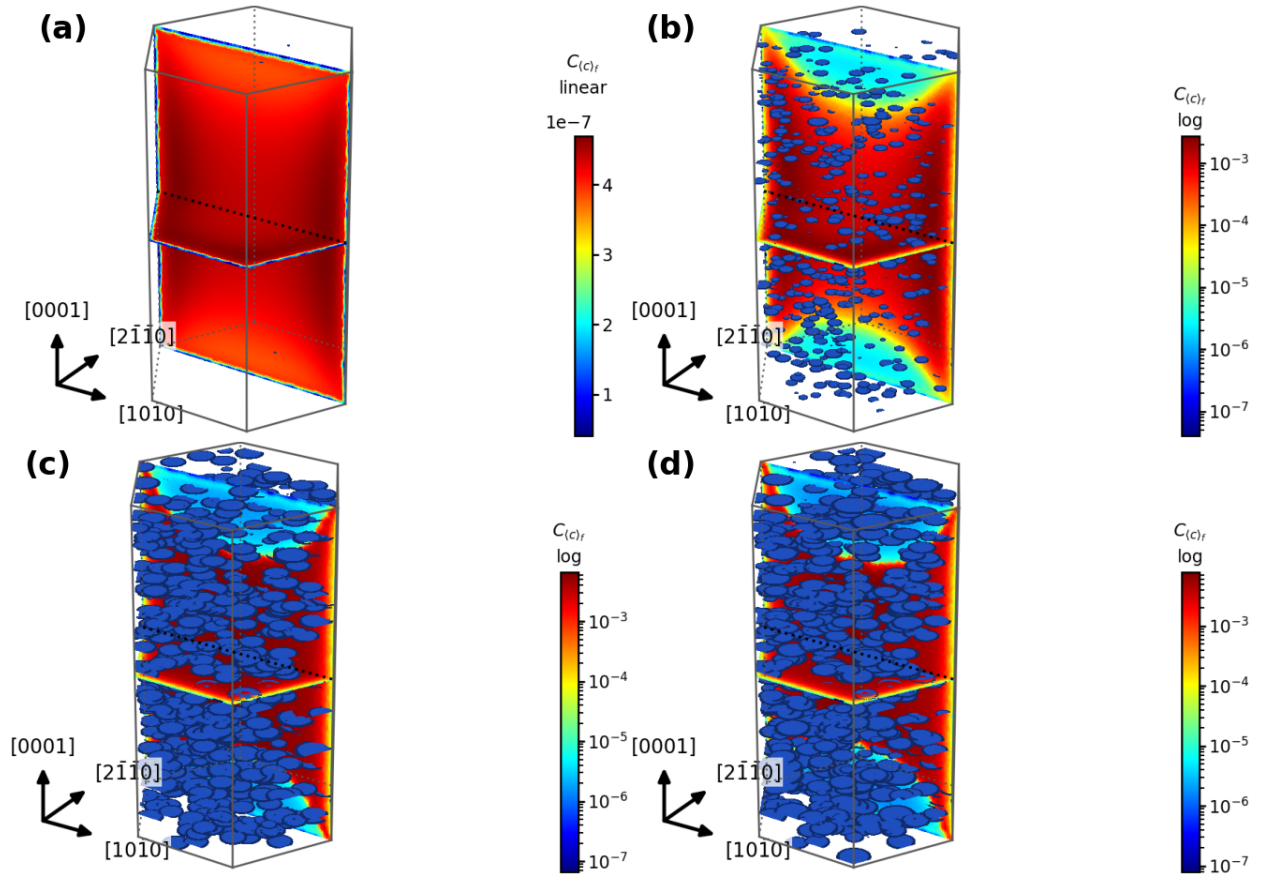


Figure 15: The discrete basal $\langle c \rangle$ population at (a) 10^{-4} , (b) 10^{-2} , (c) 1 and (d) 10 dpa, drawn at true radius over the continuum content field. Counts are 9, 858, 1031 and 1073. The loops are clipped at the crystal surface; see the text.

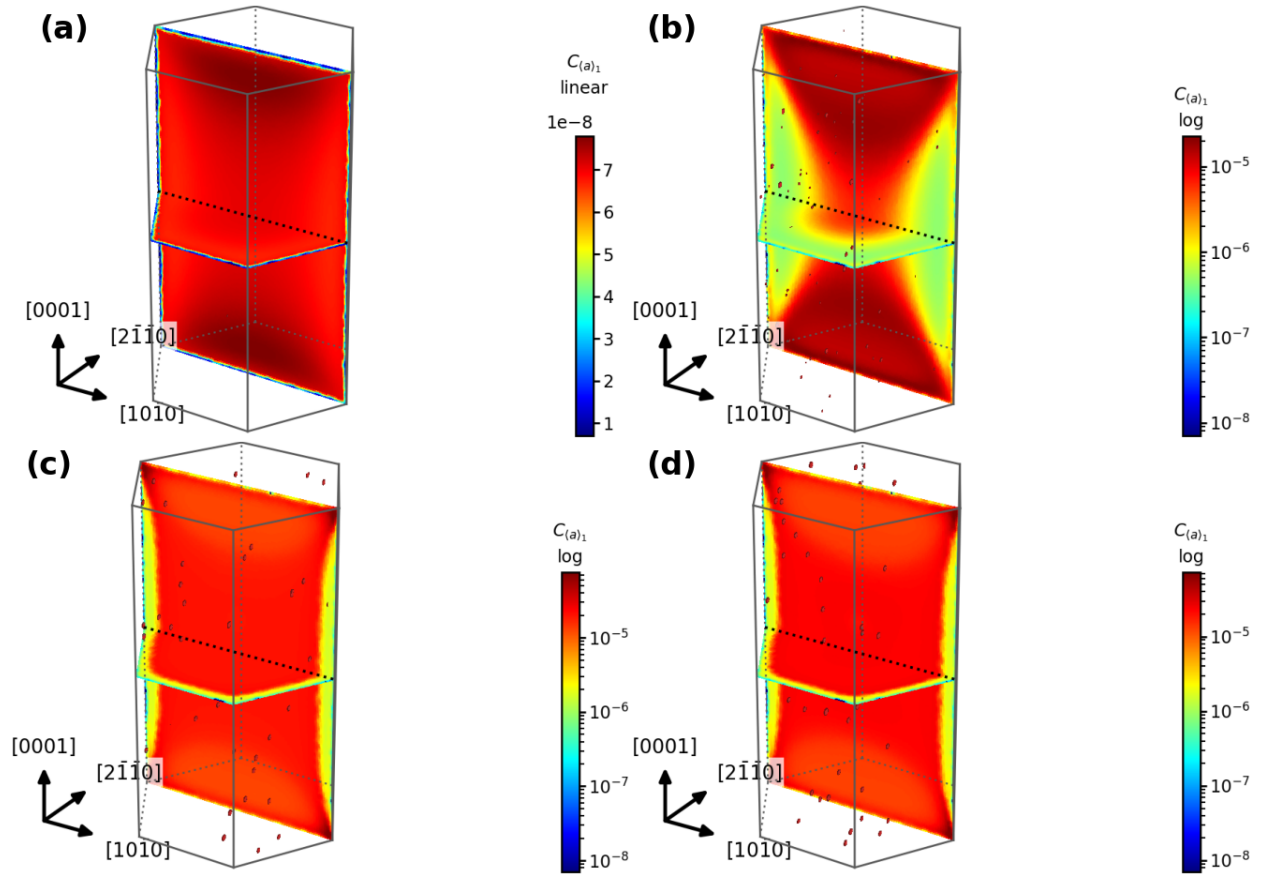


Figure 16: The discrete prismatic $\langle a \rangle_1$ population at the same four doses, (a) 10^{-4} , (b) 10^{-2} , (c) 1 and (d) 10 dpa, drawn at true radius. Counts are 2, 132, 67 and 73. At true scale these loops are 6–7 nm across in a 500 nm crystal and are correspondingly sparse; the same population drawn with exaggerated radii is legible but is not a population.

whole. This is the same geometric refusal that Section 5.3 describes, and it means the basal handoff on a 500 nm grain is only partially feasible above ~ 0.1 dpa: a discrete-dislocation solver would decline those loops, and their defects would have to remain in the continuum. *The correction to the $\langle c \rangle$ Burgers vector that made the two representations agree made this worse*, not better, because it raised the basal radius by $\sqrt{2}$ and a larger loop is harder to fit.

The conversion is where the mean field is checked against itself. At 10^{-4} dpa the population is 9 basal and 2 prismatic loops in $1.3 \times 10^8 \text{ nm}^3$ — a population so dilute that a continuum density describing it is a statement about an ensemble and not about this crystal. At 10 dpa the basal loops are 50 nm across with a mean spacing of about 50 nm, which is the opposite regime, and it is the one in which the Avrami kernels of Eq. (25) are working hardest and are least defensible. The conversion spans both, and the ledger of Eq. (41) closes on stored defects to machine precision and on loop number to the integer draw at every one of the four doses.

8. Discussion and Conclusions

8.1. What the calculation establishes

Self-consistency is achievable and is a constraint worth accepting.. Every capture efficiency in Eq. (18) and every sink strength in Eq. (14) is generated from the same diffusion tensors that transport the mobile species, through one anisotropy parameter p^m per species. The formulation therefore cannot simultaneously believe that diffusion is isotropic and that its sinks are biased by the anisotropy of that diffusion, which is a state most cluster-dynamics treatments of zirconium are free to occupy. The price is that anisotropy can no longer be absorbed by refitting a bias, and Section 6.6 shows the price being paid: the fit pushes p^v to the edge of its admissible range rather than trading it against a free efficiency. That is a feature. A model that cannot hide a deficiency in a bias factor is a model whose deficiencies are visible.

Spatial resolution changes the answer, not just its presentation.. The boundary layers of Section 7.2 are 8–46 nm wide, they order themselves by species mobility, and they narrow by a factor of two as the sink density grows — none of which a mean-field k_{gb}^2 can express. More consequentially, the loop population in the outermost shell is numerous and tiny, and it dominates every domain average: without the boundary loop channel, the domain and interior means of the $\langle c \rangle$ density differ by a factor of 873. A spatially averaged model does not have this problem because it cannot see it, and it reports the wrong number without any indication that it has.

The operator split makes the problem tractable and the cost is understood.. 7.8×10^7 stiff ordinary differential equations were integrated in 1 h 04 m on 24 cores, not because a million-equation stiff system was solved but because the frozen-mobile construction turns it into 72 928 independent 36-dimensional ones — 6.5×10^9 fewer flops per factorization — while the genuinely coupled part, having no time derivative, is solved eight times rather than thousands. The bill arrives as a first-order splitting error whose dial is the fast-solve cadence, and the two sides of the calculation cost within 7% of each other, which is the balance that cadence was chosen to produce.

Conservation closes with the boundary carried explicitly.. Both atom balances close on production to the digit, with the loop-at-boundary channel taking 5.1% of vacancy production against 0.08% of interstitial. That the channel had to be booked separately rather than left in the residual is a methodological point worth generalizing: closure by difference is not a check, because a channel omitted from it reappears silently in the remainder and nothing looks wrong.

A grain boundary absorbs loops, and the zone is a prediction.. The size-gated removal of Eq. (39) produces a denuded zone ~ 5 nm wide in which the loop density falls to the concentration floor and the mean diameter recovers monotonically from 3.5 to 58 nm. It also demonstrates something a mean-field treatment structurally cannot: a channel that acts only within a diffusion length of a surface raises the $\langle c \rangle$ density in the deep interior by a factor of 4–5, because removing the shell’s sinks changes the mobile field everywhere.

8.2. What it does not establish, and the leading open problem

The $\langle c \rangle$ population is too numerous and too small, and this is the central discrepancy.. At 573 K the interior gives N_c 9.5 times the measurement and d_c 0.53 times it. The two errors are not independent: the number balance pins the total content and the growth balance pins the size, so the density is their quotient, and a model that removes loops faster than it grows them necessarily has survivors that are too young. Every lever exercised here moves along that constraint rather than across it. Increasing coalescence brings N_c from $49\times$ to $2\times$ experiment and simultaneously collapses d_c to $0.10\times$; the small-size floor moves the prismatic families and not the basal one; gating coalescence to the large end makes matters worse by a factor of four, because the size bias is already inside the Avrami kernel and applying it twice removes disproportionate content, shrinks the family, weakens the rate and *raises* the density.

The diagnosis that survives all of this is that **a channel is missing which removes vacancy loop number without shortening the survivors' lives**. Peripheral emission shrinks a loop continuously and never removes one; coalescence removes loops in proportion to how fast they are growing; the boundary channel removes the large end but acts only near a surface, and Section 7.6 shows it does not close the interior deficit. What the data want is fewer *and larger*, and identifying the mechanism that supplies it — loop unfauling to a glissile configuration that escapes to the network, a dissolution channel for the perfect basal state, or an absorption cross-section that grows faster with size than R — is the single most valuable next piece of physics.

The basal chain is uncalibrated and switched off.. Nothing in the data supplies the collapse and unfauling barriers of Eq. (27). Switched on with placeholder rates, it inverts the family ordering: the prismatic-to-basal density ratio, which should be of order 10, becomes 0.19. That inversion is a property of the placeholders, not a prediction, and it is why the chain is inactive here. Note also that the terminal state c_p has no number-loss channel of its own beyond coalescence — it cannot unfaul further and peripheral emission acts on content only — so under a sustained vacancy supply it is bounded only by the transfer feeding it. Whether it should have a sink of its own must be settled *before* ν_{uf} is fitted; the two are not separately identifiable from a steady c_p density.

Four calibrated parameters rest on their bounds.. p^v , p^i and both Z_m^0 sit on the edge of the admissible box, so they report where the box is and not where the optimum is. The implied vacancy diffusion anisotropy, $D_{||}/D_{\text{basal}} = 16.8$, is a strong claim that the loop data cannot support on their own.

The calibration and the calculation are not a matched pair.. The parameter set was fitted in a 0-D model with no boundary, and the calculation carries a boundary loop channel that removes 5% of vacancy production.

The $\langle c \rangle$ data are thin.. Five points, two temperatures, all above 14 dpa, against four fitted parameters. The model places resolvable basal loops from 10^{-2} dpa, and nothing in the data constrains that.

No posterior was sampled.. Section 6.3 establishes that the reported set is a maximum *a posteriori* point under a stated likelihood and bounded uniform priors; it does not accompany it with a credible interval.

8.3. Path forward

Four steps, in the order their results feed one another.

1. **Refit with the boundary channel active**, on the spatially resolved model rather than in 0-D, so that the calibration and the calculation describe the same physics. This is a prerequisite for quoting any of Section 7.6 as a prediction.
2. **Constrain p^v independently**. Because p^m is the only anisotropy the formulation has, an atomistic determination enters Eq. (17) directly and converts four of the entries in Table 6 from constraints into fixed inputs. If the atomistic value is far below 1.6, the residual is a missing channel and not a transport parameter — which would be the strongest available confirmation of the diagnosis above.

3. **Run the Bayesian pipeline.** At 0.2–0.5 s per objective evaluation, a space-filling design screened by Sobol’ indices, a Gaussian-process emulator and Markov-chain sampling of Eq. (43) are all affordable, and they would replace the point estimate of Table 6 with a posterior, identify which parameters the data actually determine, and quantify how far the $\langle c \rangle$ predictions can be trusted given five measurements.
4. **Close the loop to dislocation dynamics.** The conversion of Section 5 is one-way: it exports a population and does not feed it back into a solve. Closing it — transferring a family when Eq. (35) fires, removing it from the continuum, and continuing with the loops as discrete objects whose stress fields and climb are solved — is what would make the model hybrid rather than continuum with an export. The measured obstacle is geometric and is stated in Section 7.7: a quarter of the basal loops on a 500 nm grain do not fit inside it, so either the grain must be larger or the refused population must be carried in the continuum and accounted for.

Acknowledgments

This work was performed under a Statement of Work with Fluor Marine Propulsion, LLC. The authors thank colleagues at Knolls Atomic Power Laboratory and Bettis Atomic Power Laboratory for discussions on the experimental microstructure of irradiated zirconium alloys.

Appendix A. Capture efficiencies from the diffusion tensor

The orientation factors $A_h(p)$ of Eq. (19) follow from Woo’s treatment of a dislocation loop as an absorber in an anisotropically diffusing medium [29, 30, 28], and in the tensorial form given by Po [31] the correction to the sink strength of a loop of axis α is

$$\frac{\text{tr } \mathbf{D} + \alpha^T \mathbf{D} \alpha}{4\bar{D}}, \quad (\text{A.1})$$

which is universal in loop orientation and in the degree of anisotropy. For a diagonal tensor with basal components $D_\perp$ and axial component $D_\parallel$, and with $p = (D_\parallel/D_\perp)^{1/6}$ and $\bar{D} = (\det \mathbf{D})^{1/3} = D_\perp^{2/3} D_\parallel^{1/3}$, Eq. (A.1) evaluates on a basal loop ($\alpha \parallel [0001]$) to $A_c(p) = p$ and on a prismatic loop ($\alpha \perp [0001]$) to $A_a(p) = (p + p^{-2})/2$.

The two forms differ in a way that has quantitative consequences and is easy to overlook. A_c is monotone increasing, so raising the anisotropy of a species monotonically raises its capture by basal loops. A_a has a *minimum* at $p = 2^{1/3} = 1.2599$, so raising the anisotropy from $p = 1$ first *lowers* prismatic capture. A single change in p^v therefore moves the two habits in opposite directions. Setting $p^v = 1$ where a fit had placed it at 1.179 lowers $Z_{\text{basal}}(v)$ by a factor of 0.848 while *raising* $Z_{\text{prism}}(v)$ by 1.053 — starving basal growth and accelerating prismatic shrinkage at once. Any statement about the effect of anisotropy that does not distinguish the habits is incomplete.

The co-growth condition of Section 3.5 follows directly. Both growth inequalities in Eq. (21) reduce to bounds on the single arrival ratio x , and the window between them has width $g(p^v)/g(p^i)$ with $g(p) = 2/(1 + p^{-3})$, strictly increasing — so the window is open exactly when $p^i < p^v$.

Two limits on this result should be stated. It governs the *absorption* flux only: cascade nucleation G_k is added independently and no anisotropy affects it, so the criterion does not govern net population evolution. And an isotropic calculation is not a zero of the criterion but another parameter point, so a ratio taken against one cannot test a statement about signs.

Appendix B. The moment closure, and the three rules that keep it realizable

$\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2 \geq 1$ is the Cauchy–Schwarz inequality on a non-negative measure. A state with $\Delta_k < 1$ — or with $q < 0$ outright — is therefore not an inaccurate description of a distribution; it is not the

moment triple of any distribution. Keeping the triple realizable constrains how each channel debits q , and three rules cover every channel in the model.

1. **Nucleation deposits at its declared size:** $J_{sL}^k(m_k^{\text{nuc}})^2$. This is genuine broadening, $J(m^{\text{nuc}} - \bar{m})^2 \geq 0$, and it is correct because the channel really does deposit at m_k^{nuc} .
2. **The floor current debits at $m_{\min}$:** $-m_{\min}^2 \Phi_{\min}^k$. This is a genuine statement about the lower tail.
3. **Channels with no distribution of their own are shape-preserving:** $dq = \Delta_k(2\bar{m}dc - \bar{m}^2dn)$. Annealing and coalescence carry no size spectrum, so they must not be allowed to imply one.

Rule 3 is where the sign is easily inverted, and the term is the largest in the equation. A coalescing loop does not leave the family — it *merges*. Its defects stay; only the loop-network content is lost; and fewer loops holding the same content are *larger* loops. Coalescence therefore *raises* q . Debiting it by the removed number times $\langle m^2 \rangle$, which is the natural-looking reading, inverts the sign of the term that carries 98% of the loop-number loss. The symptom in a calculation is not an error but a slow drift: on one nine-family march, q crossed zero at 1 dpa and was negative on 40% of the basal nodes at 10 dpa.

The legacy annealing surrogate cannot be made consistent, only avoided. It debits content at a *fixed* size, so the pair (dn, dc) it declares corresponds to a removed sub-population of mean $M \neq \bar{m}$. Debiting q proportionally over-removes; debiting at M — the smallest value Cauchy–Schwarz permits for the removed part — lowers Δ_k by $a(M/\bar{m} - 1)^2$ on a narrow family. No debit is realizable, because the surrogate is removing loops of a size that is not there. That is an independent argument for the peripheral-emission channel of Section Appendix C, which deletes the fitted lifetimes outright.

The gated channels — the small-size leak and the grain-boundary removal of Eq. (39) — need no rule, because removing the mass of the distribution on one side of a threshold leaves a genuine sub-measure and $\Delta_k \geq 1$ survives by construction.

Appendix C. Peripheral emission and the loop binding energy

The mean-field model represents loop annealing by two fitted lifetimes, one per character. They are replaced here by an emission channel with no fitted rate, in which a loop exchanges vacancies across its own perimeter against its own equilibrium concentration.

The binding energy of a vacancy to a loop of m defects is the difference of the loop self-energies,

$$E^b(m) = E_v^f + E_L(m - 1) - E_L(m), \quad (\text{C.1})$$

with E_L the anisotropic elastic self-energy of a circular loop of radius $R = \lambda_k \sqrt{m}$ plus, for a faulted family, its stacking-fault term:

$$E_L(m) = \frac{\mu (\mathbf{b} \cdot \mathbf{n})_k^2 R}{2(1 - \nu)} \left[\ln \frac{8R}{r_c} - 2 \right] + \pi R^2 \gamma_{\text{sf}} \delta_{k,cf}. \quad (\text{C.2})$$

The local equilibrium vacancy concentration at the loop periphery is then $c_k^{v,\text{eq}}(m) = c^{v,\text{eq}} \exp[-E^b(m)/k_B T]$, and the emission current is the same capture expression run backwards against it,

$$S_{vL}^k = \Pi_{sL}^k \bar{D}^v Z_{vk,v} c_k^{v,\text{eq}}(\bar{m}_{sL}^k), \quad (\text{C.3})$$

which enters Eq. (10) through $\mathbf{S}_{vL}$ and Eq. (33) with the opposite sign.

Three properties recommend this over a fitted lifetime. *One expression serves both characters:* the same mechanism shrinks a vacancy loop, by emitting the vacancies it holds, and grows an interstitial one, by emitting vacancies that recombine with its interstitials, so no separate interstitial-loop annealing model is needed. *The climb threshold emerges rather than being imposed:* growth requires the arriving flux to exceed Eq. (C.3), which reproduces the classical critical radius without it appearing as a parameter. And *it removes content, not loops*, which is physically correct — emission shrinks a loop continuously — but has a consequence that must be recognized: after this replacement, coalescence becomes the *only* sink on vacancy loop number. That is stated in Section 8 as the leading open problem, and it is a consequence of doing the emission physics correctly rather than an artifact of it.

Appendix D. State-vector layout and the conservation accumulators

The integrated state at each node is a single flat vector whose width depends on the model controls. The layout is fixed and every index keeps its meaning as the controls add fields:

Indices	Contents	Count
0–3	mobile: c^v, c^i, c^{2i}, c^{3i}	4
4–7	number of families 0–3	4
8–11	content of families 0–3	4
12–17	accumulators: production, recombination, sink absorption, each per character	6
18	network dislocation density ρ_N	1
19–23	number of families 4–8	5
24–28	content of families 4–8	5
29–37	second content q of all nine families	9
38–39	grain-boundary loop ledger, per character	2

Widths are therefore 19 (four families, two moments), 29 (nine families, two moments), 38 (nine families, three moments) and 40 with the boundary ledger. Two properties of this layout are worth recording because both were established by their failure modes.

The appended families are appended, not interleaved. Every index below 19 means what it meant in the four-family model, so a state written by one configuration is readable by another up to its own width. The corollary is that a narrower state is always *accepted* by a wider solver, the appended slots defaulting to zero — which is exactly why a mismatch between the width a driver allocates and the width the model requires is silent rather than fatal, and why the checkpoint fingerprint records the actual width.

The block written to the spatial code is ordered the other way. It is moment-major — all nine numbers, then all nine contents, then all nine q — so the bridge between the two is a scatter and not a copy. Exactly one routine knows that mapping; performing it at a call site is what once put an 18-wide block into an 8-wide slice.

The accumulators are per-interval; the ledger is cumulative. The six conservation accumulators are reset at the start of each dose interval, so a stored snapshot holds one interval’s production and must be summed to give a running total. The two grain-boundary accumulators are never reset and are already cumulative. Differencing the wrong one against the wrong other gives a channel that appears to carry 6600 times total production, or none at all, depending on which way the error runs.

References

- [1] B. Adamson, R. E. Coleman, C. M. Griffiths, Irradiation effects in zirconium alloys, Tech. Rep. IAEA-TECDOC-2020, International Atomic Energy Agency, Vienna, comprehensive IAEA review: irradiation growth, loop data, SPP evolution, void data, dose/temperature dependences for BWR, PWR, and CANDU Zircalloys. (2019).
- [2] C. Lemaignan, A. T. Motta, Zirconium alloys in nuclear applications, in: R. W. Cahn, P. Haasen, E. J. Kramer (Eds.), Materials Science and Technology: A Comprehensive Treatment, Volume 10B, VCH, 1994, pp. 1–51.
- [3] D. O. Northwood, Irradiation damage in zirconium and its alloys, Atomic Energy Review 15 (4) (1977) 547–610.
- [4] M. Griffiths, A review of microstructure evolution in zirconium alloys during irradiation, Journal of Nuclear Materials 159 (1988) 190–218, review: $\langle c \rangle$ -component loops and voids in Zr; formation mechanism; temperature window 625–775 K for voids; He role in cavity stabilization; irradiation growth mechanisms; SPP influence. doi:10.1016/0022-3115(88)90029-5.

- [5] F. Onimus, J.-L. Béchade, Radiation effects in zirconium alloys, in: R. J. M. Konings (Ed.), *Comprehensive Nuclear Materials*, Elsevier, 2012, Ch. 4.01, pp. 1–31. doi:10.1016/B978-0-08-056033-5.00064-4.
- [6] M. N. Cinbiz, D. A. Koss, A. T. Motta, The influence of stress state on the reorientation of hydrides in a zirconium alloy, *Journal of Nuclear Materials* 477 (2016) 157–164. doi:10.1016/j.jnucmat.2016.05.013.
- [7] N. M. Ghoniem, A dynamic rate theory for the response of metals during steady state and pulsed irradiation, Ph.d. thesis (nuclear engineering), University of Wisconsin–Madison (1977).
- [8] N. M. Ghoniem, D. D. Cho, The simultaneous clustering of point defects during irradiation, *Physica Status Solidi (a)* 54 (1979) 171–178. doi:10.1002/pssa.2210540121.
- [9] N. M. Ghoniem, S. Sharafat, A numerical solution to the Fokker–Planck equation describing the evolution of the interstitial loop microstructure during irradiation, *Journal of Nuclear Materials* 92 (1980) 121–135. doi:10.1016/0022-3115(80)90148-8.
- [10] M. R. Hayns, The transient stages of damage accumulation in irradiated materials with point defect clusters, *Journal of Nuclear Materials* 59 (1976) 175–187. doi:10.1016/0022-3115(76)90163-5.
- [11] M. Kiritani, Analysis of the clustering process of supersaturated lattice vacancies, *Journal of the Physical Society of Japan* 35 (1973) 95–107. doi:10.1143/JPSJ.35.95.
- [12] A. D. Brailsford, R. Bullough, The rate theory of swelling due to void growth in irradiated metals, *Journal of Nuclear Materials* 44 (2) (1972) 121–135. doi:10.1016/0022-3115(72)90091-8.
- [13] H. Wiedersich, On the theory of void formation during irradiation, *Radiation Effects* 12 (1972) 111–125. doi:10.1080/00337577208231128.
- [14] L. K. Mansur, Void swelling in metals and alloys under irradiation: an assessment of the theory, *Nuclear Technology* 40 (1) (1978) 5–34. doi:10.13182/NT78-2.
- [15] D. J. Bacon, F. Gao, Y. N. Osetsky, The primary damage state in fcc, bcc and hcp metals as seen in molecular dynamics simulations, *Journal of Nuclear Materials* 276 (2000) 1–12. doi:10.1016/S0022-3115(99)00165-8.
- [16] J. Marian, C. S. Becquart, C. Domain, S. L. Dudarev, M. R. Gilbert, R. J. Kurtz, D. R. Mason, K. Nordlund, A. E. Sand, L. L. Snead, T. Suzudo, B. D. Wirth, Recent advances in modeling and simulation of the exposure and response of tungsten to fusion energy conditions, *Nuclear Fusion* 57 (2017) 092008. doi:10.1088/1741-4326/aa5e8d.
- [17] S. I. Golubov, A. M. Ovcharenko, A. V. Barashev, B. N. Singh, Grouping method for the approximate solution of a kinetic equation describing the evolution of point-defect clusters, *Philosophical Magazine A* 81 (2001) 643–658. doi:10.1080/01418610108212164.
- [18] A. Holt, R. Microstructure of Zr-2.5Nb pressure tubes, *Journal of Nuclear Materials* 159 (1988) 310–338, zr-2.5Nb CANDU pressure tube microstructure under neutron irradiation; He bubbles at grain boundaries ($\sim 5 \times 10^{20} \text{ m}^{-3}$, $\sim 1.5 \text{ nm}$); β -Zr decomposition; irradiation growth and creep. doi:10.1016/0022-3115(88)90030-1.
- [19] M. Christensen, W. Wolf, C. Freeman, E. Wimmer, R. B. Adamson, L. Hallstadius, P. E. Cantonwine, E. V. Mader, Diffusion of point defects, nucleation of dislocation loops, and effect of hydrogen in hcp-Zr: Ab initio and classical simulations, *Journal of Nuclear Materials* 460 (2015) 82–96. doi:10.1016/j.jnucmat.2015.02.013.
- [20] A. Patra, D. R. Trinkle, First-principles vacancy diffusion and Onsager transport coefficients in Zr, *Computational Materials Science* 170 (2019) 109176. doi:10.1016/j.commatsci.2019.109176.

- [21] A. A. Kohnert, B. D. Wirth, L. Capolungo, Modeling microstructural evolution in irradiated materials with cluster dynamics methods: a review, *Computational Materials Science* 149 (2018) 442–459. doi:10.1016/j.commatsci.2018.02.049.
- [22] M. Maron, Y. Li, I. Lalani, K. Baker, B. Ramirez Flores, T. Black, J. Hollenbeck, N. Ghoniem, G. Po, Spatially-resolved cluster dynamics modeling of irradiation growth, *International Journal of Plasticity* 181 (2024) 103989. doi:10.1016/j.ijplas.2024.103989.
- [23] F. Christien, A. Barbu, Cluster dynamics modelling of irradiation growth of zirconium single crystals, *Journal of Nuclear Materials* 393 (1) (2009) 153–161.
- [24] A. V. Barashev, S. I. Golubov, R. E. Stoller, Theoretical investigation of microstructure evolution and deformation of zirconium under neutron irradiation, *Journal of Nuclear Materials* 461 (2015) 85–94.
- [25] Y. Li, M. Maron, K. Baker, B. Ramirez Flores, T. Black, J. Hollenbeck, I. Lalani, N. Ghoniem, G. Po, Coupled cluster and dislocation dynamics modeling of microstructure evolution in irradiated materials, *Journal of the Mechanics and Physics of Solids* 206 (2025) 106366. doi:10.1016/j.jmps.2025.106366.
- [26] C. Ji, Y. Cui, Y. Li, N. Ghoniem, A concurrent irradiation-mechanics multiscale coupling model, *Journal of the Mechanics and Physics of Solids* 167 (2022) 105005. doi:10.1016/j.jmps.2022.105005.
- [27] Y. Li, G. Po, N. Ghoniem, Coupled cluster-dislocation dynamics of irradiation-induced defects, *Materialia* 14 (2020) 100891.
- [28] C. Woo, Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion, *Journal of Nuclear Materials* 159 (1988) 237–256.
- [29] C. H. Woo, The sink strength of a dislocation loop in the effective medium approximation, *Journal of Nuclear Materials* 98 (3) (1981) 279–294. doi:10.1016/0022-3115(81)90154-9.
- [30] C. H. Woo, Intrinsic bias differential between vacancy loops and interstitial loops, *Journal of Nuclear Materials* 107 (1) (1982) 20–30. doi:10.1016/0022-3115(82)90554-3.
- [31] G. Po, A tensorial derivation of the DAD effect, Tech. rep., Department of Mechanical and Aerospace Engineering, University of Miami, internal technical note (2026).
- [32] A. Jostsons, P. Kelly, R. Blake, The nature of dislocation loops in neutron irradiated zirconium, *Journal of Nuclear Materials* 66 (3) (1977) 236–256.
- [33] M. Griffiths, A review of microstructure evolution in zirconium alloys during irradiation, *Journal of Nuclear Materials* 159 (1988) 190–218.
- [34] G. Carpenter, J. Watters, A study of electron irradiation damage in zirconium using a high voltage electron microscope, *Journal of Nuclear Materials* 96 (3) (1981) 213–226.
- [35] L. Tournadre, F. Onimus, J.-L. Béchade, D. Gilbon, J.-M. Cloué, J.-P. Mardon, X. Feaugas, O. Toader, C. Bachelet, Experimental study of the nucleation and growth of c-component loops under charged particle irradiations of recrystallized zircaloy-4, *Journal of nuclear materials* 425 (1-3) (2012) 76–82.
- [36] A. Harte, D. Jädnäs, M. Topping, P. Frankel, C. Race, J. Romero, L. Hallstadius, E. C. Darby, M. Preuss, The effect of matrix chemistry on dislocation evolution in an irradiated zr alloy, *Acta Materialia* 130 (2017) 69–82.
- [37] P. Kelly, R. Blake, The characterization of dislocation loops in neutron irradiated zirconium, *Philosophical Magazine* 28 (2) (1973) 415–426.

- [38] S. I. Choi, J. H. Kim, Radiation-induced dislocation and growth behavior of zirconium and zirconium alloys – a review, *Nuclear Engineering and Technology* 45 (3) (2013) 385–392. doi:10.5516/NET.07.2013.035.
- [39] G. J. C. Carpenter, J. F. Watters, R. W. Gilbert, Dislocations generated by neutron irradiation in zirconium, *Journal of Nuclear Materials* 159 (1988) 86–100.
- [40] X. Yi, M. L. Jenkins, M. Briceno, S. G. Roberts, Z. Zhou, M. A. Kirk, In situ study of self-ion irradiation damage in W and Zr at room temperature, *Philosophical Magazine* 93 (14) (2013) 1715–1738. doi:10.1080/14786435.2012.754110.
- [41] S. Doriot, B. Verhaeghe, J.-L. Béchade, D. Menut, D. Gilbon, J.-P. Mardon, B. Cazalis, J.-L. Bernier, Microstructural evolution of M5 alloy under PWR conditions, *Journal of ASTM International* 12 (2015).
- [42] R. Holt, R. Gilbert, $\langle c \rangle$ component dislocations in annealed zircaloy irradiated at about 570 k, *Journal of Nuclear Materials* 137 (3) (1986) 185–189.
- [43] R. A. Holt, A. R. Causey, Irradiation creep and growth of Zr-2.5Nb pressure tubes, *Journal of Nuclear Materials* 209 (1993) 221–231.
- [44] J. F. March-Rico, B. D. Wirth, Stress states and point defect capture radii of dislocation a-loops and c-loops in alpha-zirconium, *Journal of Nuclear Materials* 587 (2023) 154752. doi:10.1016/j.jnucmat.2023.154752.
- [45] J. F. March-Rico, S. Blondel, B. D. Wirth, Atomistically-informed cluster dynamics modeling of concurrent interstitial and vacancy a-loop evolution in irradiated alpha-zirconium, *Acta Materialia* 288 (2025) 120777. doi:10.1016/j.actamat.2025.120777.
- [46] S.-M. Liu, I. J. Beyerlein, W.-Z. Han, Two-dimensional vacancy platelets as precursors for basal dislocation loops in hexagonal zirconium, *Nature Communications* 11 (1) (2020) 5766.
- [47] C. Varvenne, O. Mackain, E. Clouet, Vacancy clustering in zirconium: An atomic-scale study, *Acta materialia* 78 (2014) 65–77.
- [48] B. Singh, On the influence of grain boundaries on void growth, *Philosophical Magazine* 28 (6) (1973) 1409–1413.
- [49] X.-M. Bai, A. F. Voter, R. G. Hoagland, M. Nastasi, B. P. Uberuaga, Efficient annealing of radiation damage near grain boundaries via interstitial emission, *Science* 327 (5973) (2010) 1631–1634.
- [50] C. Sun, M. Song, K. Yu, Y. Chen, M. Kirk, M. Li, H. Wang, X. Zhang, In situ evidence of defect cluster absorption by grain boundaries in kr ion irradiated nanocrystalline ni, *Metallurgical and Materials Transactions A* 44 (2013) 1966–1974.
- [51] G. Cheng, W. Xu, Y. Wang, A. Misra, Y. Zhu, Grain size effect on radiation tolerance of nanocrystalline mo, *Scripta Materialia* 123 (2016) 90–94.